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THE COMPLETE WORLD OF EDUCATION

11/28/2025

Cloud Security Fundamentals

Practical File ENCS-353

B. Tech CSE (Spec. Cyber Security)

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K.R. MANGALAM UNIVERSITY

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PRACTICAL-1

OBJECTIVE - TO CREATE EC2 INSTANCE

Steps to Create an EC2 Instance

1. Login to AWS

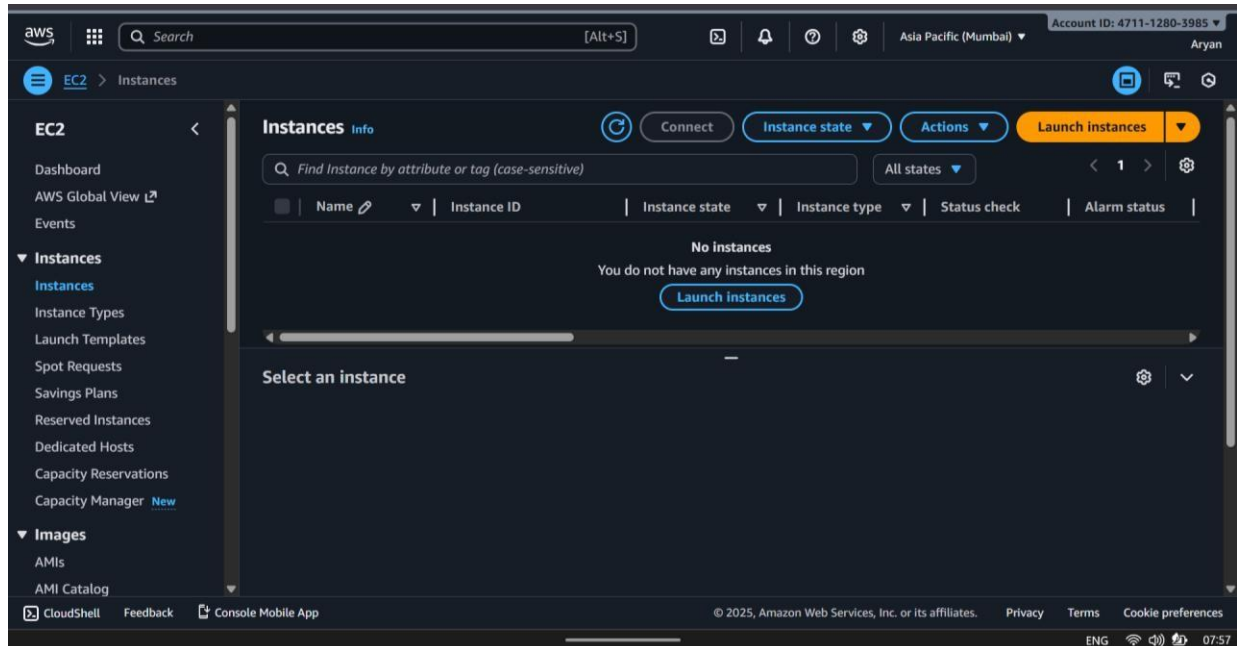
- Go to AWS Management Console.
- Sign in with your credentials.

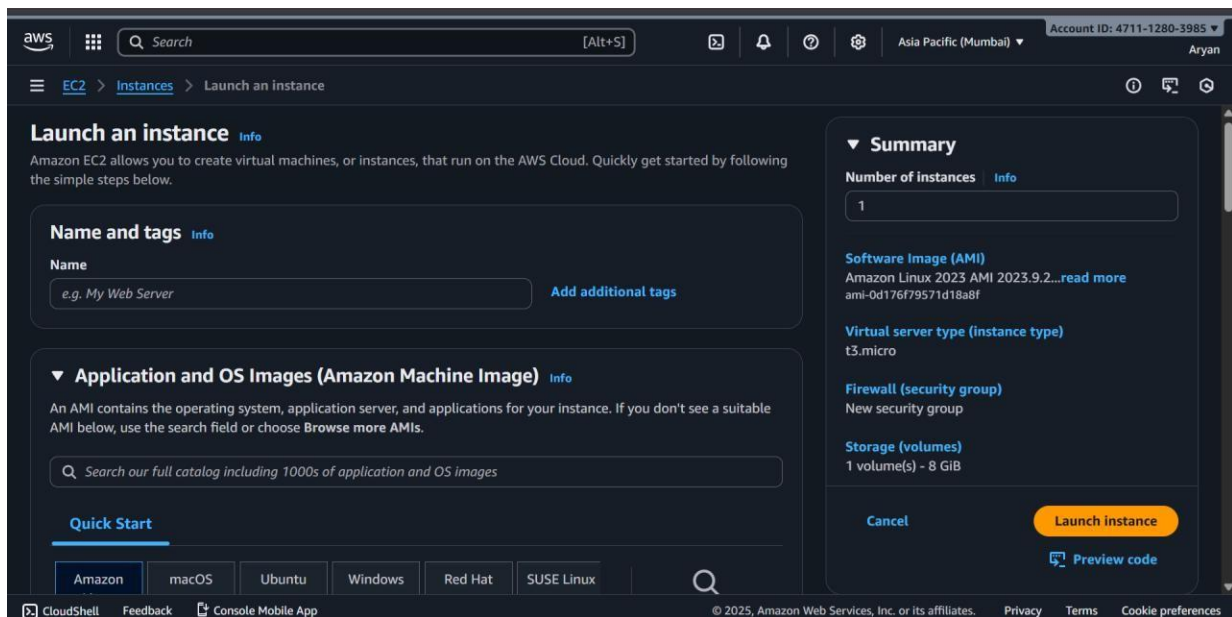
2. Navigate to EC2

- In the search bar, type **EC2** and select it.

3. Launch Instance

- Click **Instances** → **Launch Instance**.
- Give your instance a **name**.





4. Choose Amazon Machine Image (AMI)

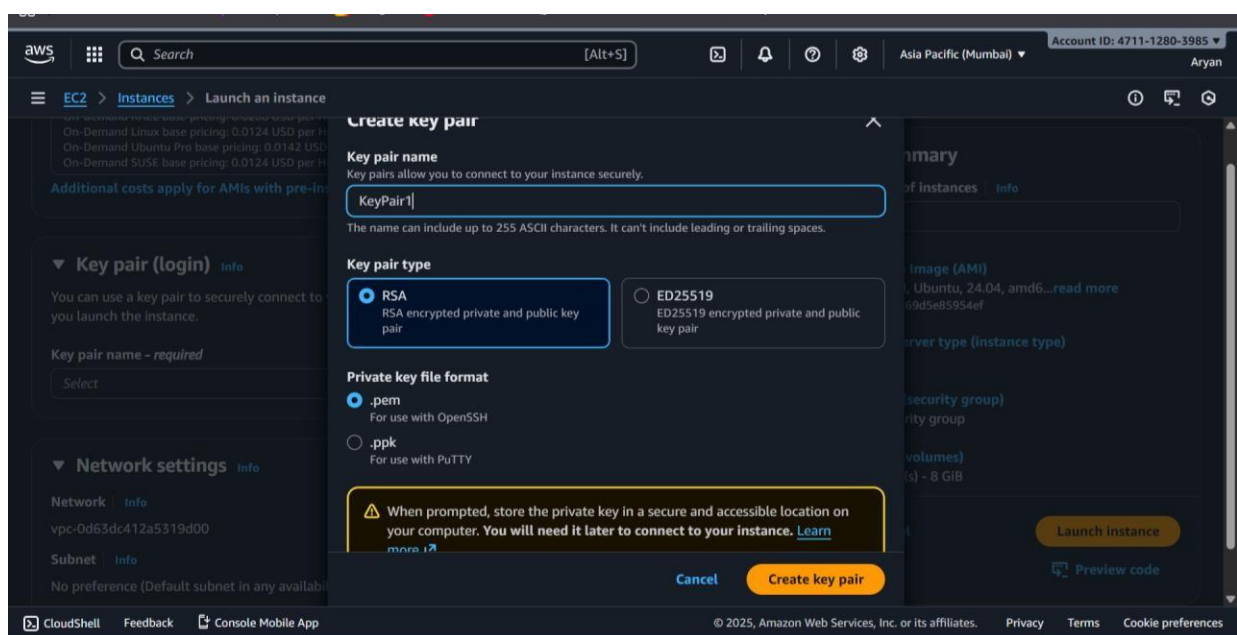
- Select an OS image (e.g., Amazon Linux 2, Ubuntu, Windows).
- For beginners, choose **Amazon Linux 2 (Free tier eligible)**.
- I chose **Ubuntu**.

5. Choose Instance Type

- Select **t2.micro** (free tier eligible).

6. Configure Key Pair

- Create a new key pair (RSA) and download the **.pem file**.



This will be used later to connect to your instance.

7. Configure Network & Security

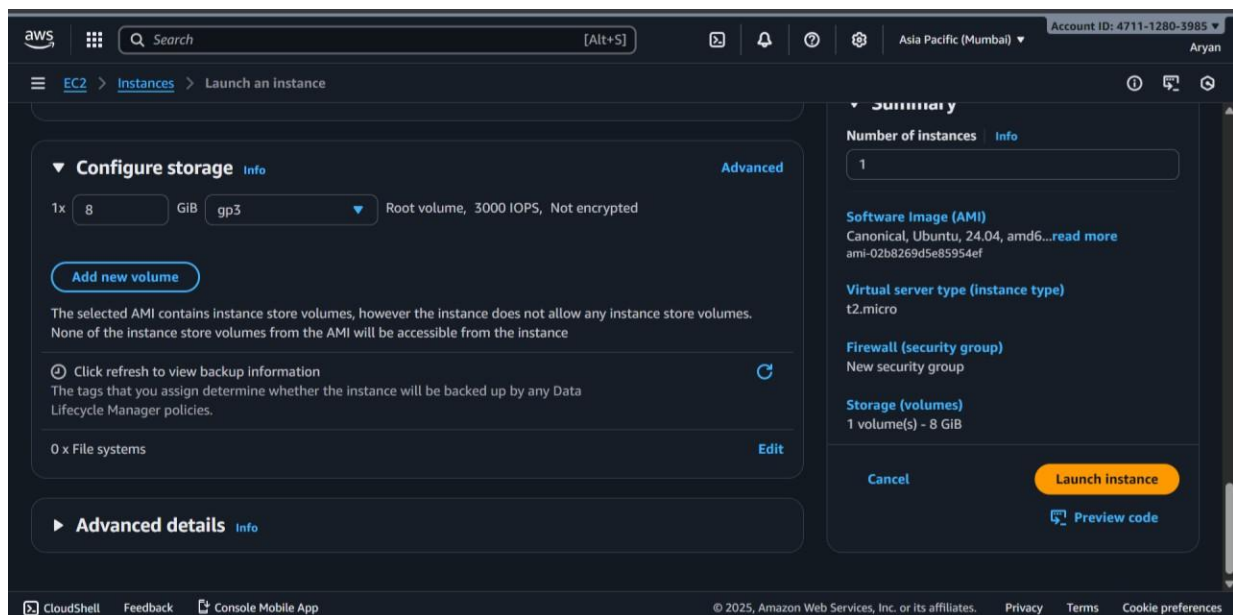
- Leave default **VPC and subnet** (unless you want custom networking).
- Under **Security group**, allow:
 - **SSH (port 22)** for Linux
 - **RDP (port 3389)** for Windows
 - **HTTP/HTTPS** if you're hosting a website.

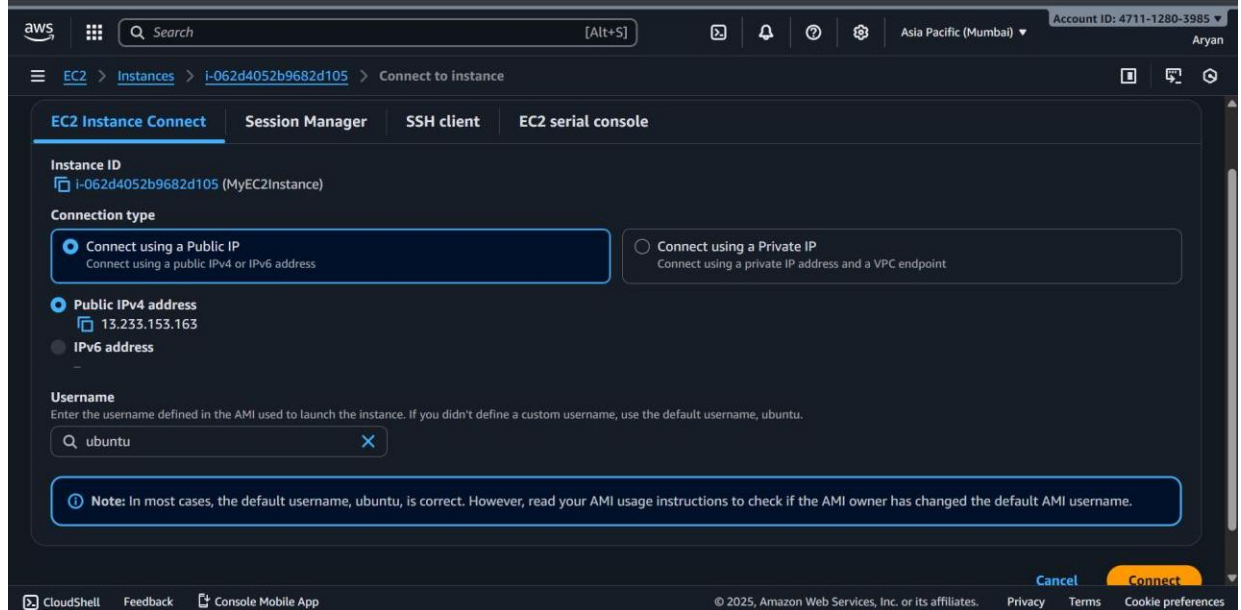
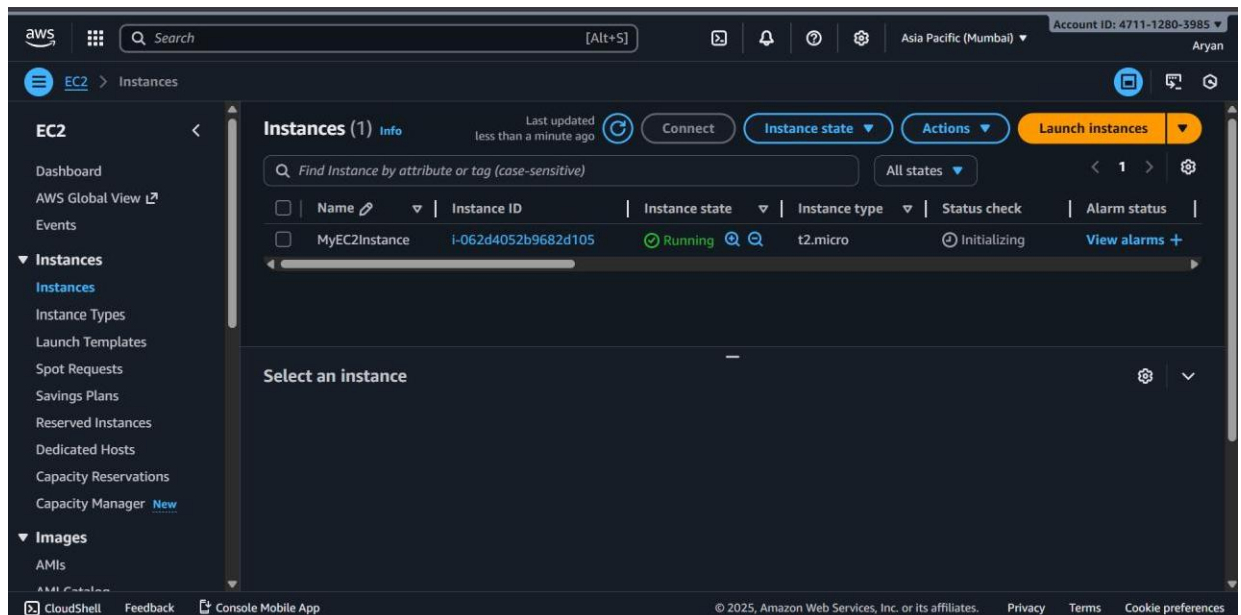
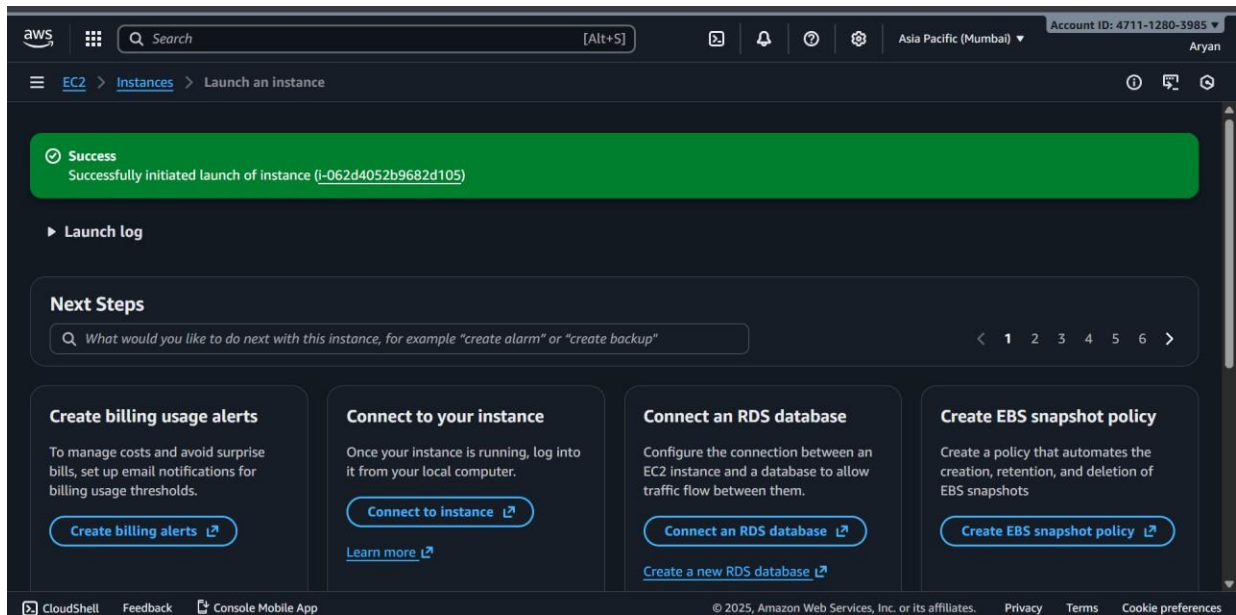
8. Configure Storage

- Default 8 GB is fine (increase if needed).

9. Launch Instance

- Click **Launch Instance**.
- AWS will start your instance.





aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

* Support: <https://ubuntu.com/pro>

System information as of Fri Nov 28 02:41:10 UTC 2025

System load: 0.31

Usage of /: 25.8% of 6.71GB

Memory usage: 21%

Swap usage: 0%

Processes: 110

Users logged in: 0

IPv4 address for enX0: 172.31.6.85

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.

To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

ubuntu@ip-172-31-6-85:~\$ sudo su
root@ip-172-31-6-85:/home/ubuntu#

i-062d4052b9682d105 (MyEC2Instance)

PublicIPs: 13.233.153.163 PrivateIPs: 172.31.6.85

CloudShell

Feedback

Console Mobile App

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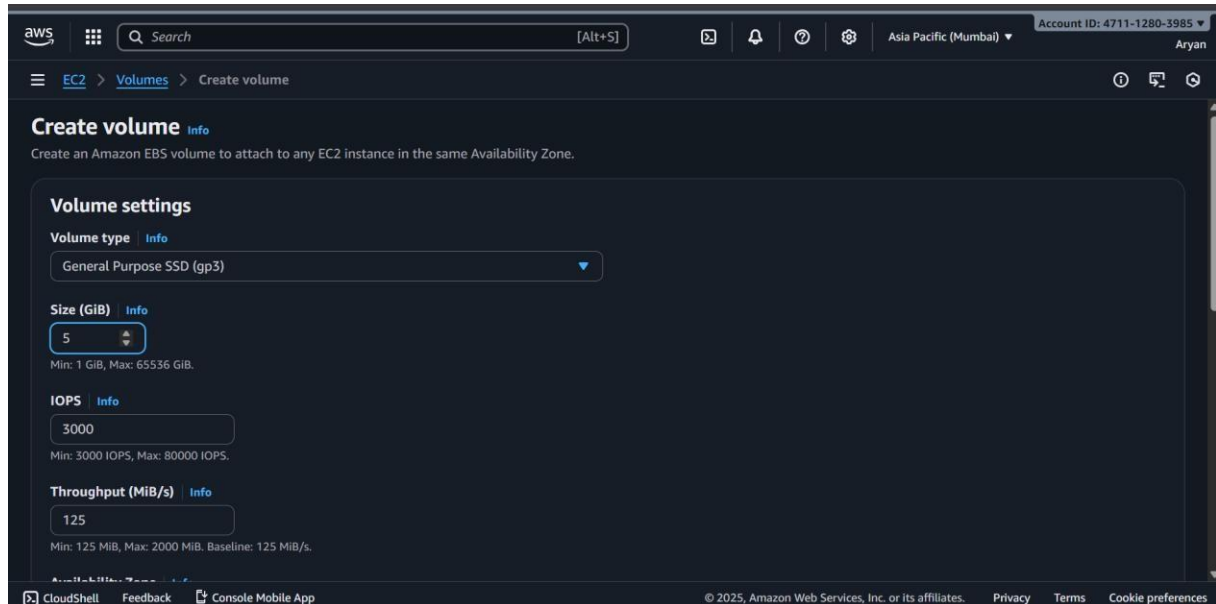
Terms

Cookie preferences

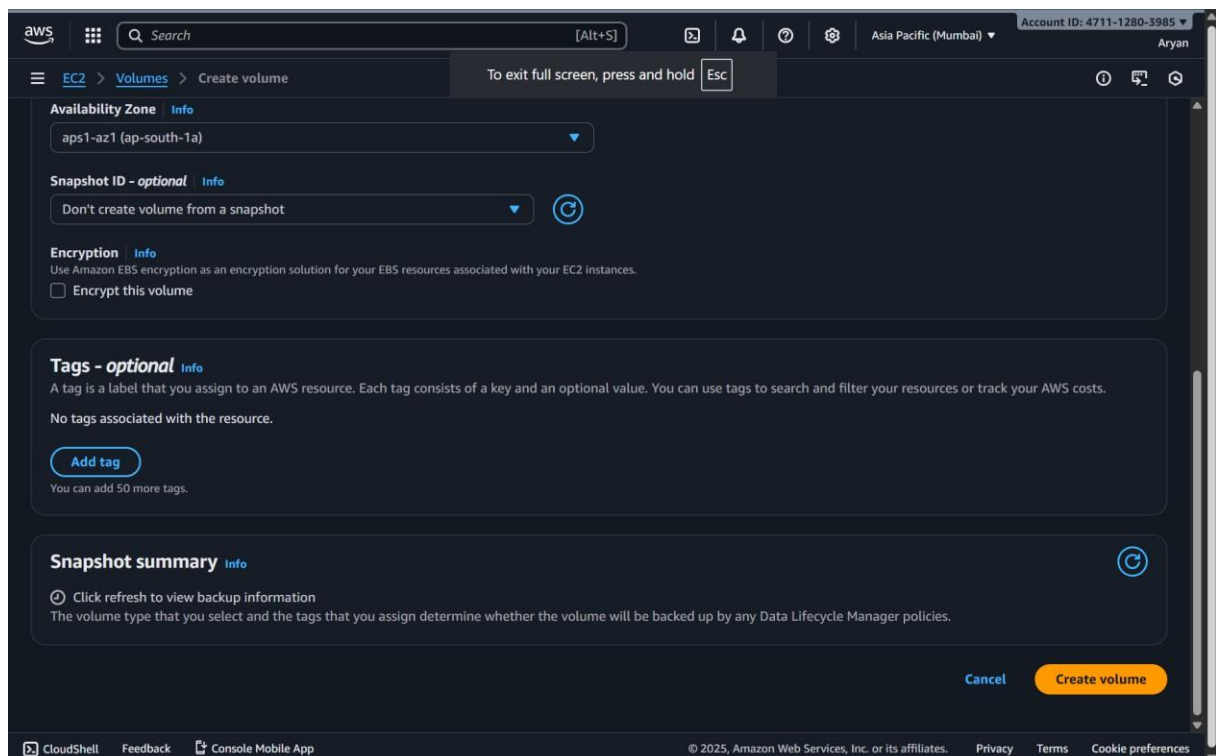
Practical:- 2

Objective:- Create Amazon Elastic Block Store (EBS) volumes in AWS

Step1:- Search EC2 and scroll to volumes.



Step 2:- Click on create volume and choose the settings as per the requirements. (here we have chosen gp3 -> size=5 and rest are default.



Step 3:- Keep the availability zone as the required zone in other activities too.

The screenshot displays the AWS Management Console interface. The top section shows the 'Attach volume' page for volume `vol-05cd0b9b6dabcb4d1` in the `ap-south-1a` availability zone, attached to instance `i-00122ecfde1397844`. Below this, the 'Volumes (2)' page is shown, listing two volumes: `vol-0e434cccb9e8cbf92` (8 GiB) and `vol-0c08d2f9a05266840` (5 GiB). The volume `vol-0e434cccb9e8cbf92` is highlighted in red, indicating it is newly created. The 'Snapshot summary' section shows 0 recently backed up volumes out of 0 total volumes.

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
`vol-05cd0b9b6dabcb4d1`

Availability Zone
`ap-south-1a`

Instance [Info](#)
`i-00122ecfde1397844`
(mywebserver) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
`/dev/sd[...]`

Recommended device names for Linux: `/dev/xvda` for root volume, `/dev/sd[f-p]` for data volumes.

Volumes (2) [Info](#) Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Table:

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID
<code>vol-0e434cccb9e8cbf92</code>	<code>vol-0e434cccb9e8cbf92</code>	gp3	8 GiB	3000	125	<code>snap-0a07784...</code>
<code>vol-0c08d2f9a05266840</code>	<code>vol-0c08d2f9a05266840</code>	gp3	5 GiB	3000	125	-

Fault tolerance for all volumes in this Region

Snapshot summary Last updated on Fri, Nov 28, 2025, 08:17:19 AM (GMT+05:30)

Recently backed up volumes / Total # volumes
0 / 0

Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | [Create policy](#)

Newly created volume is marked by the red colour.

PRACTICAL -3

OBJECTIVE- CREATE AN S3 BUCKET IN AWS

1. Log in to AWS Management Console

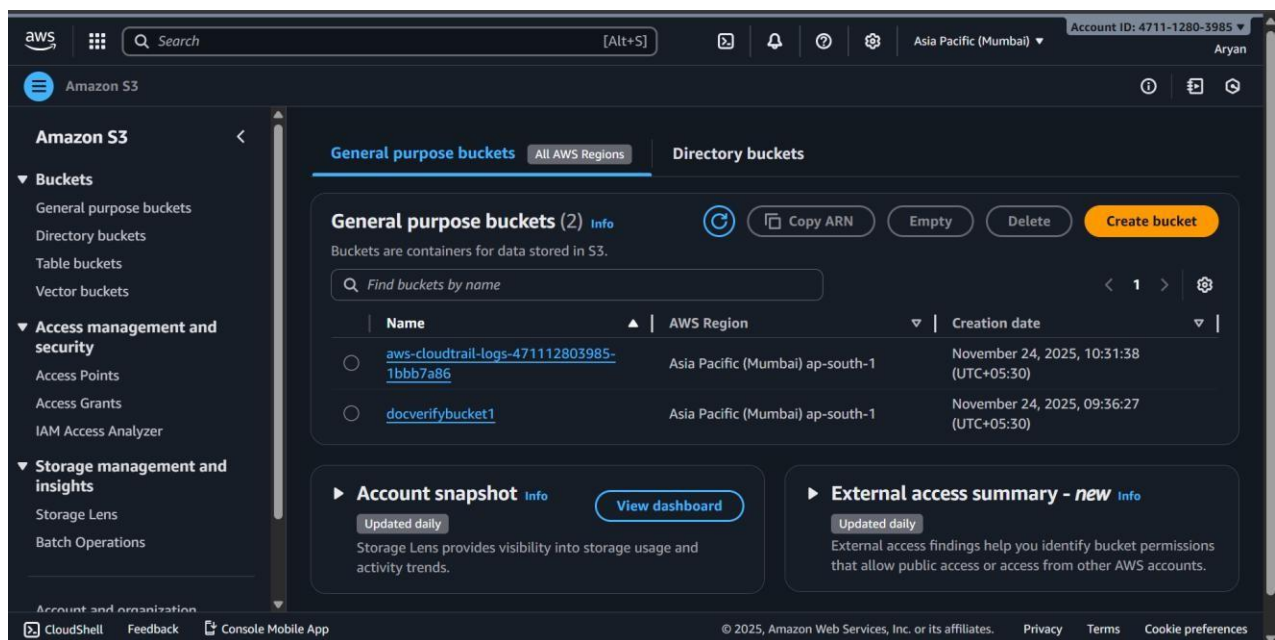
- Go to AWS Console.
- Sign in with your AWS account credentials.

2. Navigate to S3

- In the search bar at the top, type **S3**.
- Click on **S3 (Scalable Storage in the Cloud)**.

3. Create a New Bucket

- Click on “**Create bucket**” button.



4. Configure Bucket Settings

- **Bucket name:** Enter a unique name (e.g., my-first-s3-bucket-2025).

Bucket names must be globally unique across AWS.

- **AWS Region:** Select a region (choose one closest to your users for better performance).

The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The breadcrumb navigation indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'Info' link. Below this, a note states 'Buckets are containers for data stored in S3.' The 'General configuration' section is expanded, showing the 'AWS Region' as 'Asia Pacific (Mumbai) ap-south-1'. Under 'Bucket type', the 'General purpose' option is selected, with a description: 'Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.' The 'Directory' option is also visible but unselected. The 'Bucket name' field is labeled 'MyFirstBucket' and has a note below it stating: 'Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more'. At the bottom of the configuration section, there is a 'Choose bucket' button. The footer of the console shows 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services, Inc. or its affiliates.

5. Set Bucket Options

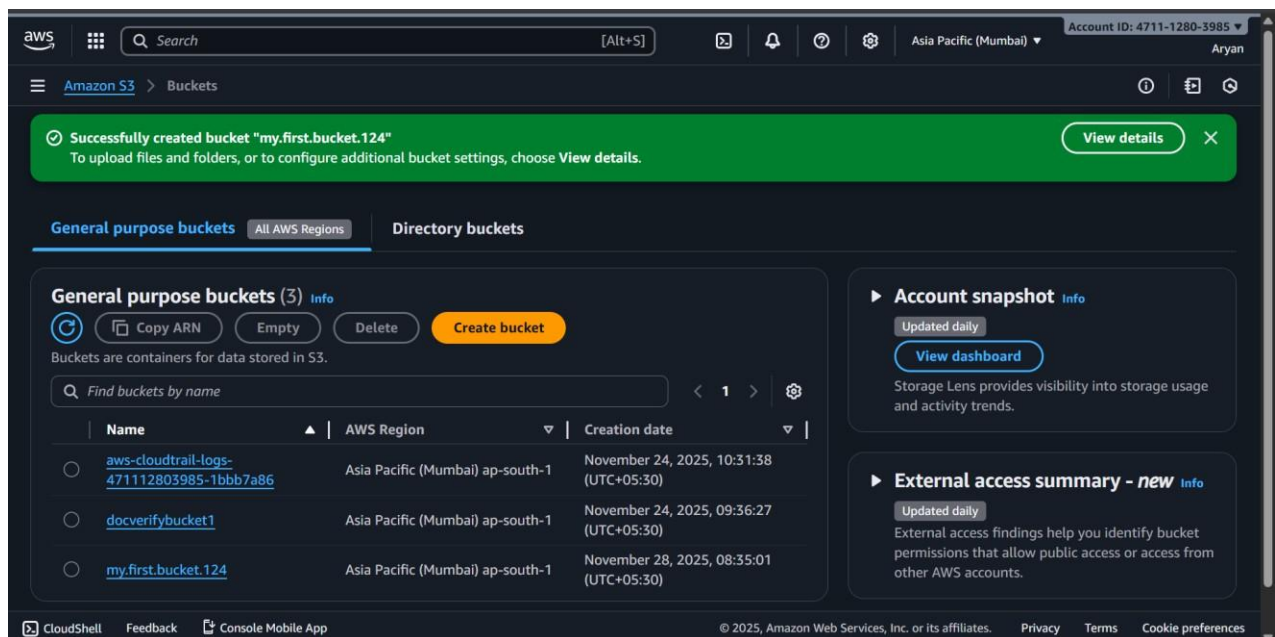
- **Object Ownership:**
 - Choose **ACLs disabled (recommended)** for most use cases.
- **Block Public Access:**
 - By default, all public access is blocked (recommended for private buckets).
 - If you need a public bucket (for hosting static websites), uncheck the option and acknowledge the warning.

6. Configure Bucket Settings (Optional)

- **Versioning:** Enable if you want to keep multiple versions of an object.
- **Encryption:** Enable default encryption if required.
- **Tags:** Add key-value tags if you want to manage cost tracking or organization.

7. Review and Create

- Review your settings.
- Click **Create bucket**.



8. Upload Objects (Optional)

- After creation, open the bucket.
- Click **Upload** → **Add files**.
- Select your file(s), then click **Upload**.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

Amazon S3

Buckets

my.first.bucket.124

my.first.bucket.124

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name

Type

Last modified

Size

Storage class

No objects

You don't have any objects in this bucket.

Upload

CloudShell

Feedback

Console Mobile App

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PRACTICAL-4

OBJECTIVE – CREATE AN AWS LAMBDA FUNCTION

1. Log in to AWS Management Console

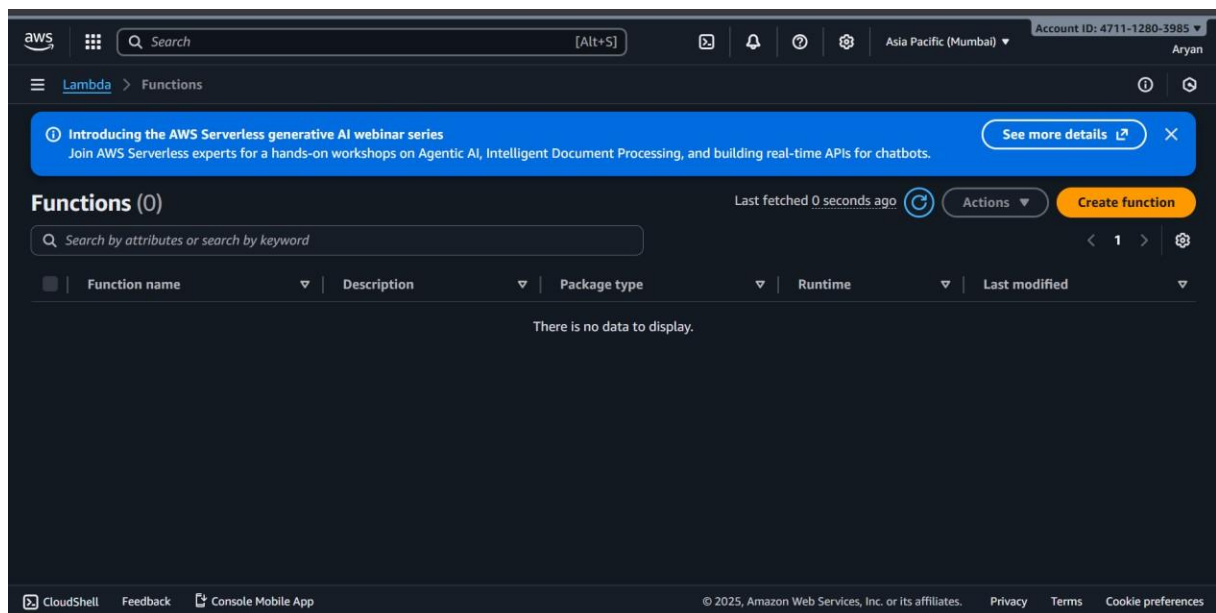
- Go to AWS Console.
- Sign in with your credentials.

2. Navigate to AWS Lambda

- In the search bar, type **Lambda**.
- Click on **Lambda** service.

3. Create a New Lambda Function

- Click **Create function**.



4. Choose a Creation Method

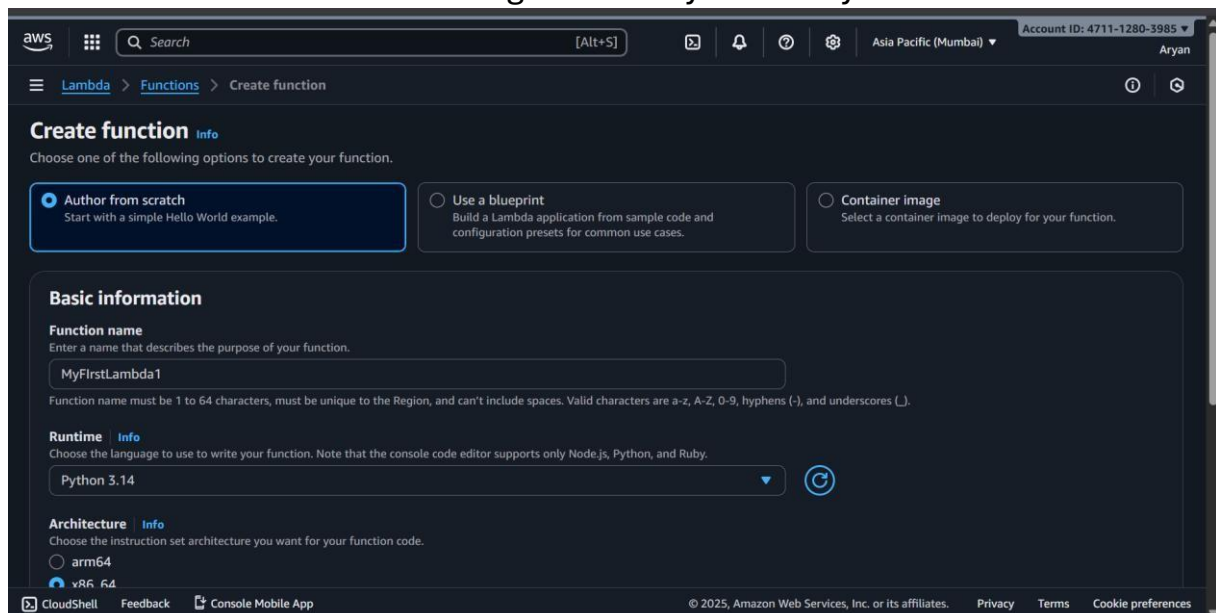
You'll see 3 options:

1. **Author from scratch** → (most common, start fresh).
2. **Use a blueprint** → predefined templates.
3. **Container image** → deploy code as Docker container. Select

Author from scratch.

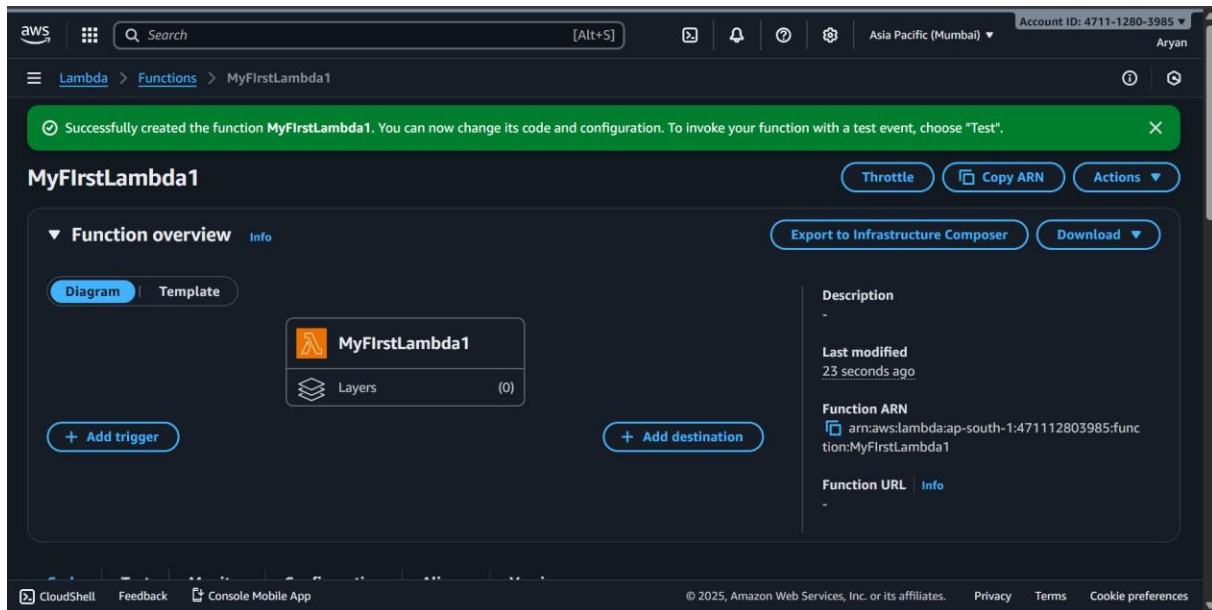
5. Configure Basic Settings

- **Function name:** Enter a unique name (e.g., MyFirstLambda).
- **Runtime:** Choose a runtime (Node.js, Python, Java, Go, etc. → example: **Python 3.9**).
- **Permissions (Execution Role):**
 - **Create a new role with basic Lambda permissions** (recommended if you're new).
 - Or choose an existing IAM role if you already have one.



6. Click Create Function

- Wait a few seconds while AWS provisions the function.

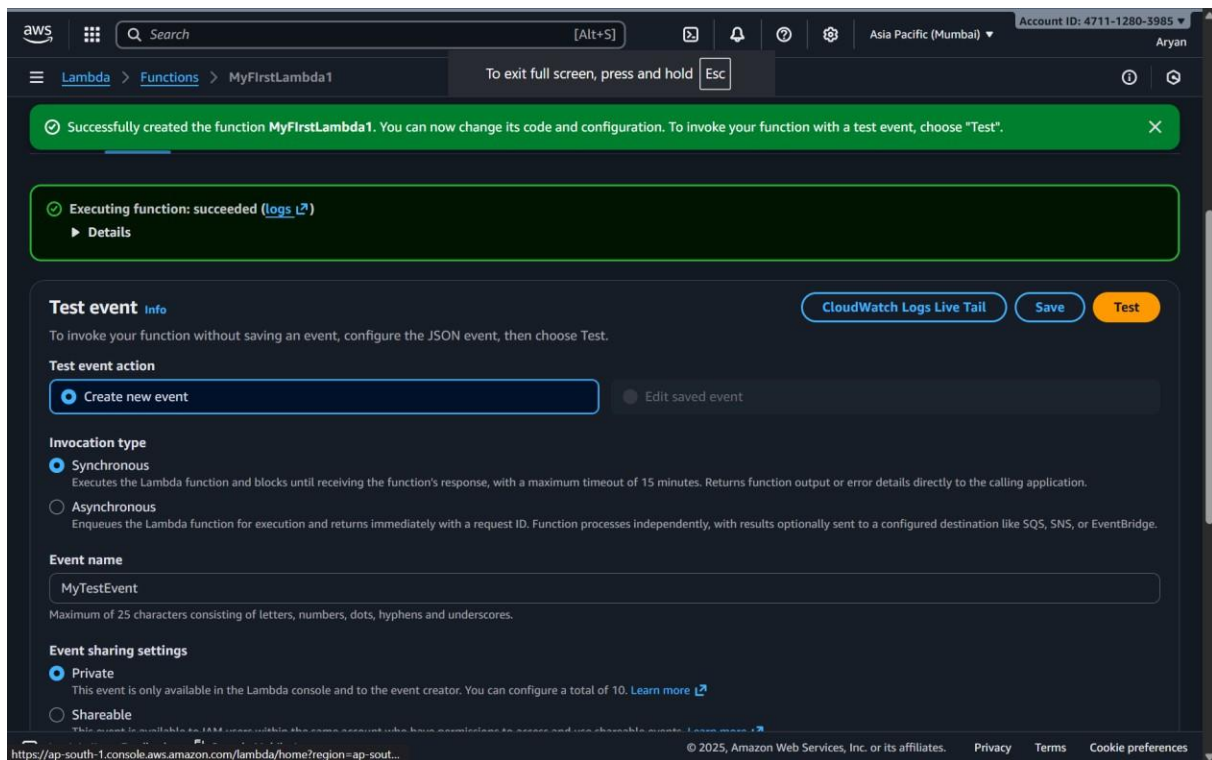


7. Add Your Code

- In the **Function code** section, you can:
 - Write inline code in the editor.
 - Or upload a .zip file.
 - Or use **Amazon S3** (if your code is stored there).

Example default code in Python:

```
def lambda_handler(event, context):  
    return {  
        'statusCode': 200,  
        'body': 'Hello from Lambda!'  
    }
```



8. Configure Test Event

- Click **Test** → **Configure test event**.
- Give it a name (e.g., TestEvent).
- Keep the default event JSON or modify as needed.
- Save.

9. Run the Function

- Click **Test** again.
- Check the execution results (output, logs, and status).

10. (Optional) Add a Trigger

- You can connect Lambda to services like:
 - API Gateway (for REST APIs)
 - S3 (trigger on file uploads)
 - DynamoDB (trigger on database changes)
 - EventBridge (scheduled events)

aws

Search

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Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

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Lambda

Add triggers

Trigger configuration

Info

S3

aws asynchronous storage

Bucket

Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

s3/my.first.bucket.124

X

🔄

Bucket region: ap-south-1

Event types

Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

All object create events X

Prefix - optional

Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.

e.g. images/

CloudShell

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Account ID: 4711-1280-3985

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Lambda

Functions

MyFirstLambda1

MyFirstLambda1

Throttle

Copy ARN

Actions

The trigger my.first.bucket.124 was successfully added to function MyFirstLambda1.

X

Function overview

Info

Diagram

Template

MyFirstLambda1

Layers

(0)

S3

+ Add trigger

+ Add destination

Export to Infrastructure Composer

Download

Description

-

Last modified

15 minutes ago

Function ARN

arn:aws:lambda:ap-south-1:471112803985:function:MyFirstLambda1

Function URL

Info

CloudShell

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Account ID: 4711-1280-3985

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Upload succeeded

For more information, see the [Files and folders](#) table.

Summary

Destination

s3://my.first.bucket.124

Succeeded

1 file, 33.0 B (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 33.0 B)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
Random Test File.txt	-	text/plain	33.0 B	Succeeded	-

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Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

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Lambda > Functions > MyFirstLambda1

Successfully updated the function MyFirstLambda1.

Code source

Info

Open in Visual Studio Code

Upload from

EXPLORER

MYFIRSTLAMBDA1

lambda_function.py

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [NONE SELECTED]

Create new test event

lambda_function.py

```
1
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
7         'body': json.dumps('Hello from Lambda!')
8     }
9 print("Hello World")
```

CloudShell

Feedback

Console Mobile App

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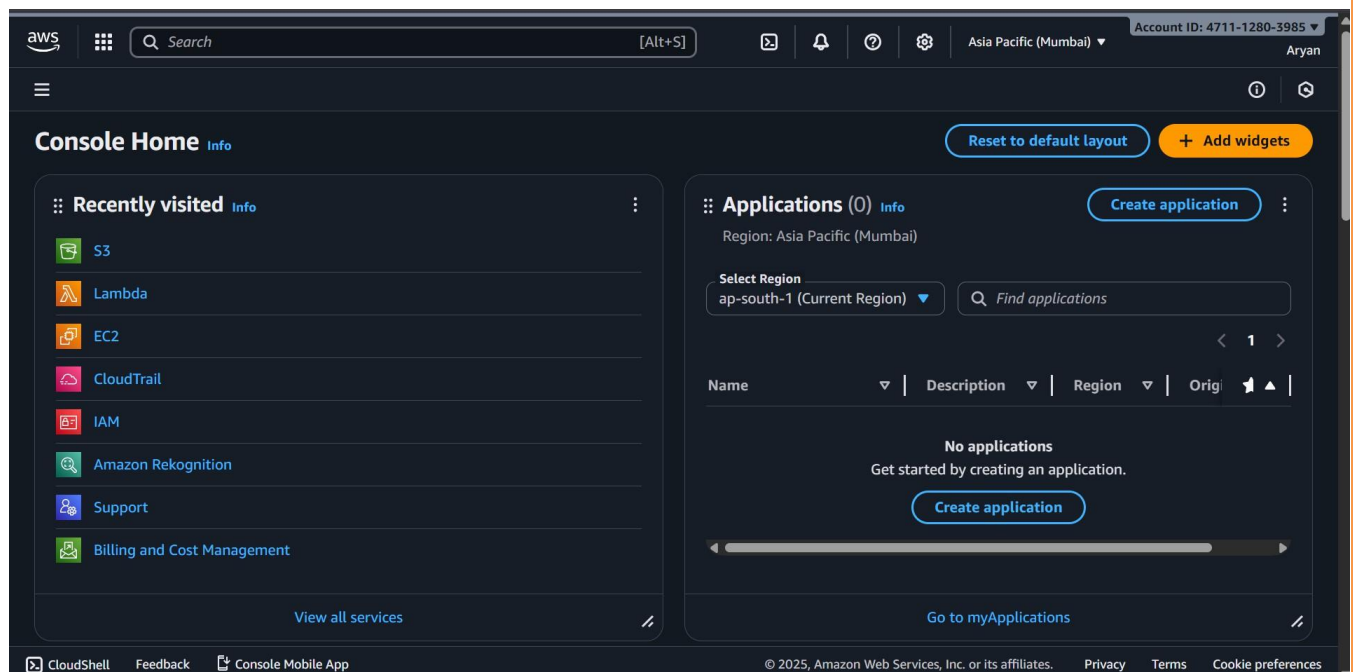
Cookie preferences

Practical:-5

Objective:- Creating an IAM (Identity and Access Management) user in AWS is to provide secure and controlled access to AWS resources for individuals or applications without using the root account.

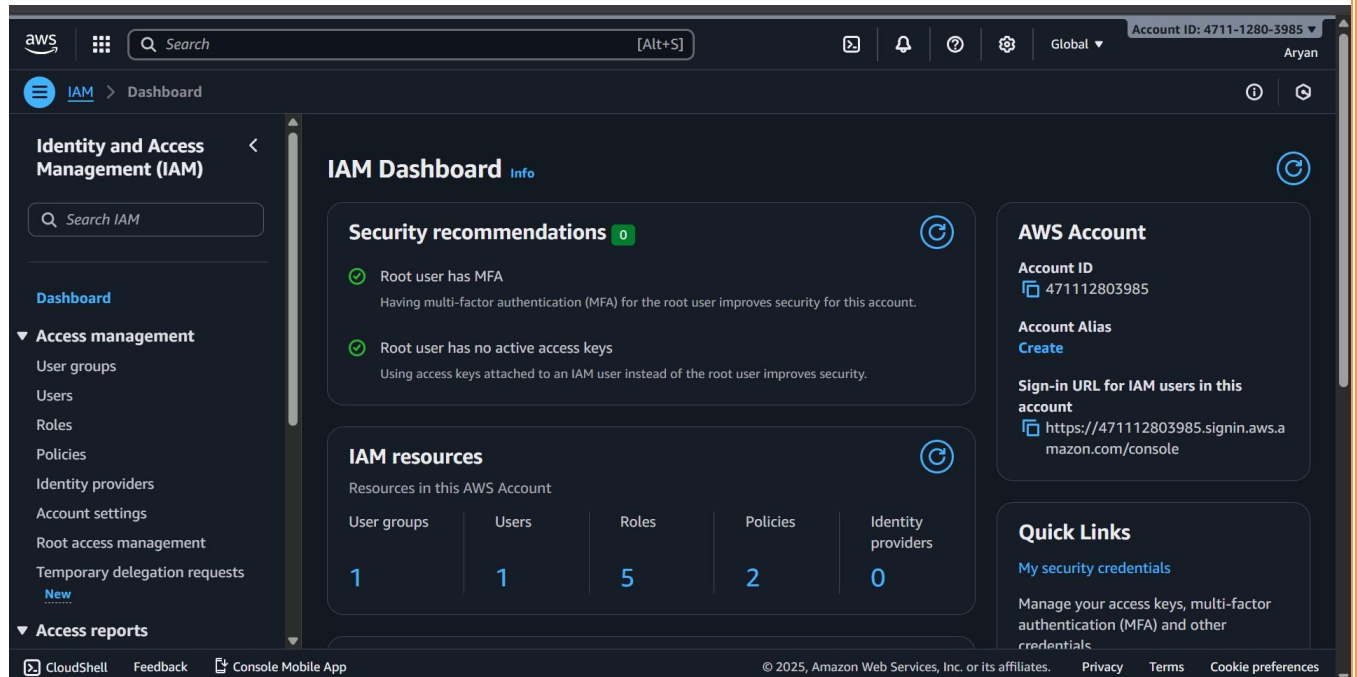
Step 1: Sign in to AWS Console

Log in to your **AWS Management Console** using your **root account** or an **IAM user** who has **Administrator privileges**.



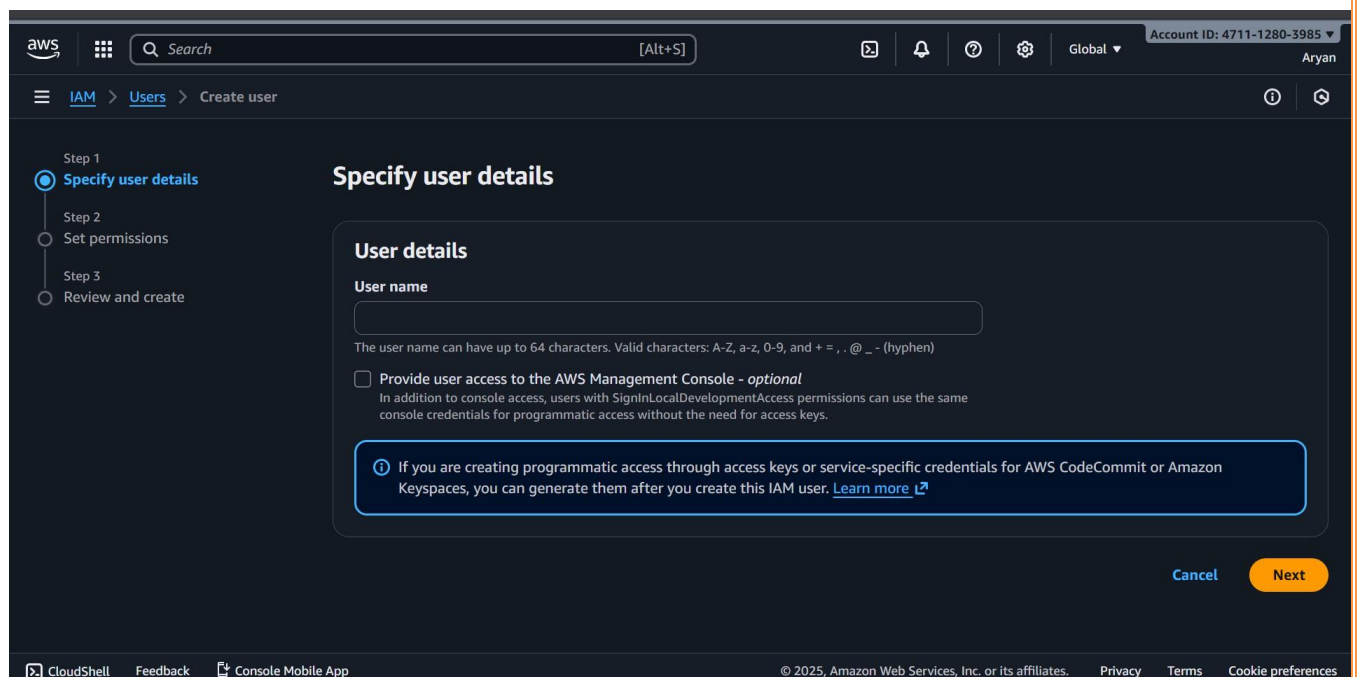
Step 2: Open IAM Service

In the search bar at the top, type **IAM**. Select **IAM (Identity and Access Management)**.



Step 3: Go to Users Section

In the left sidebar, click on **Users**. You will see a list of all existing users. Click **“Create user”**



to add a new one.

Step 4: Enter User Details

Enter Username. Choose the type of access. Review access, if the user needs to log

in to the AWS Console. Click **Next**.

aws

Search

[Alt+S]

Account ID: 4711-1280-3985

Aryan

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Specify user details

User details

User name

Aryan_IAM_User_1

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional

In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ Autogenerated password

You can view the password after you create the user.

☒ Custom password

Enter a custom password for the user.

☐ Show password

☒ Users must create a new password at next sign-in - Recommended

Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

CloudShell

Feedback

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Step 5: Set Permissions

Choose **Attach policies directly** → assign permissions manually (e.g., AmazonS3FullAccess, AdministratorAccess, etc.). Then

click **Next**.

aws

Search

[Alt+S]

Account ID: 4711-1280-3985

Aryan

IAM > Users > Create user

more

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1425)

Create policy

Choose one or more policies to attach to your new user.

Filter by Type

Search

All types

< 1 2 3 4 5 6 7 ... 72 >

☐	Policy name	Type	Attached entities
☐	AccessAnalyzerServiceRol...	AWS managed	0
☐	AdministratorAccess	AWS managed - job function	0
☐	AdministratorAccess-Amp...	AWS managed	0

CloudShell

Feedback

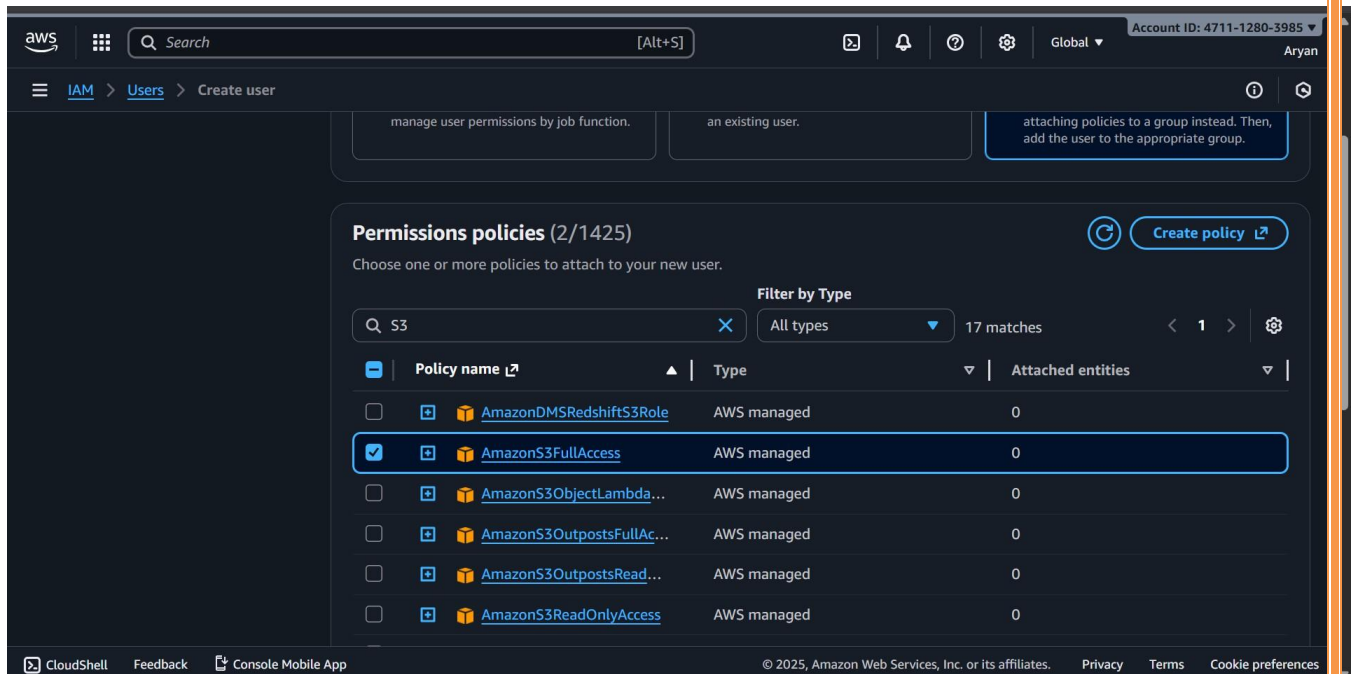
Console Mobile App

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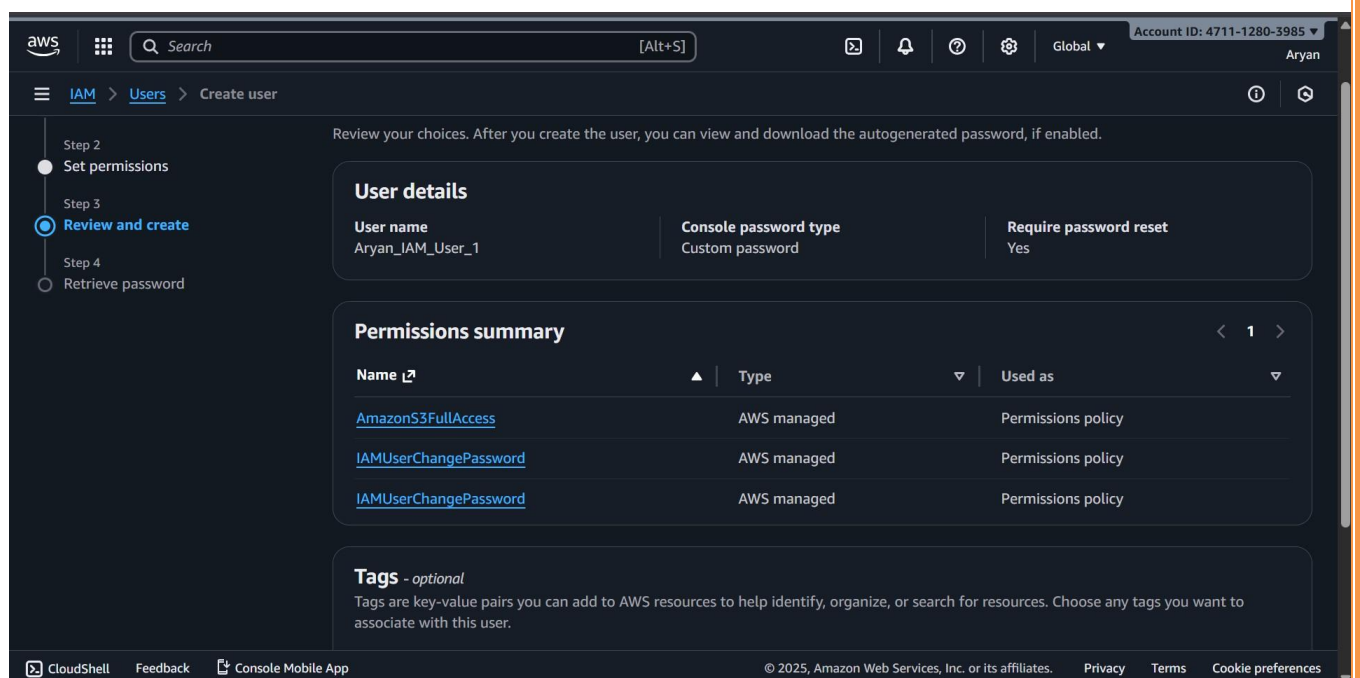
Terms

Cookie preferences



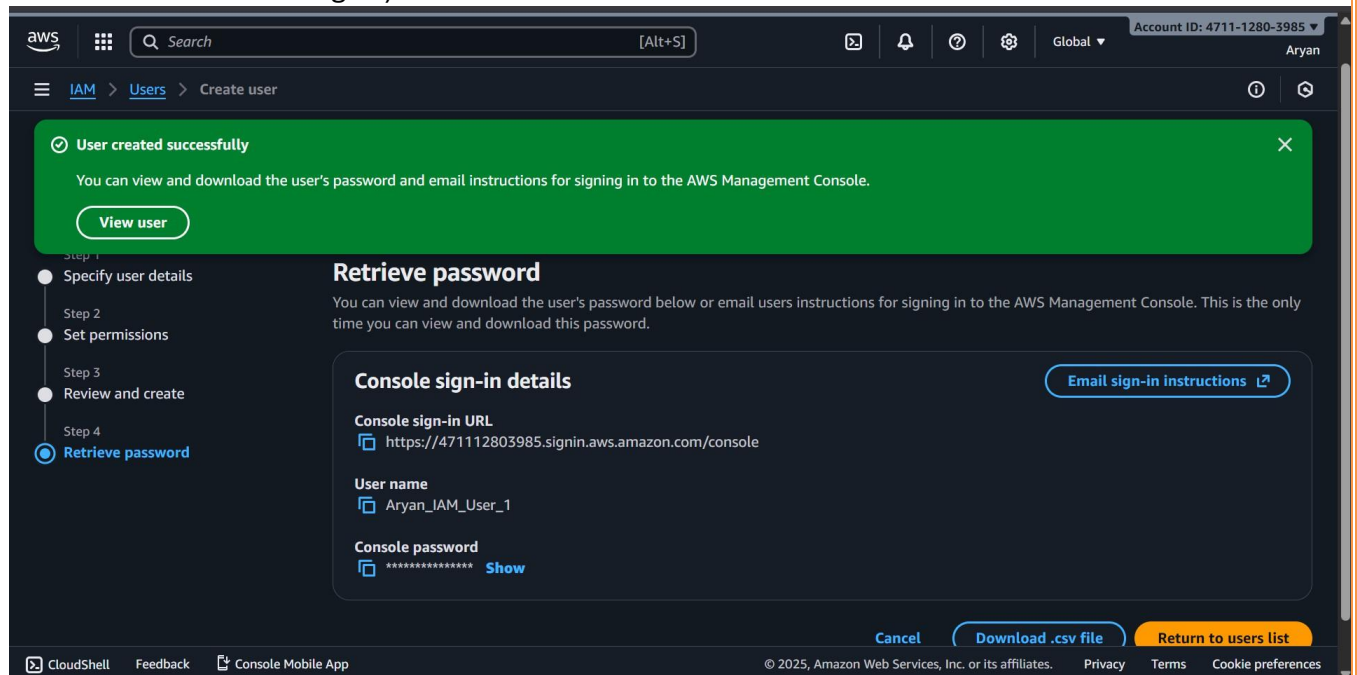
Step 6: Review and Create

Review all details carefully. Click **Create user**.



Step 7: Save Login Details

- Once the user is created, AWS will show:
 - **User ARN (Amazon Resource Name)**
 - **Console login link**
 - **Password or Access key/Secret key** (Download the .csvfile — it won't be shown again).

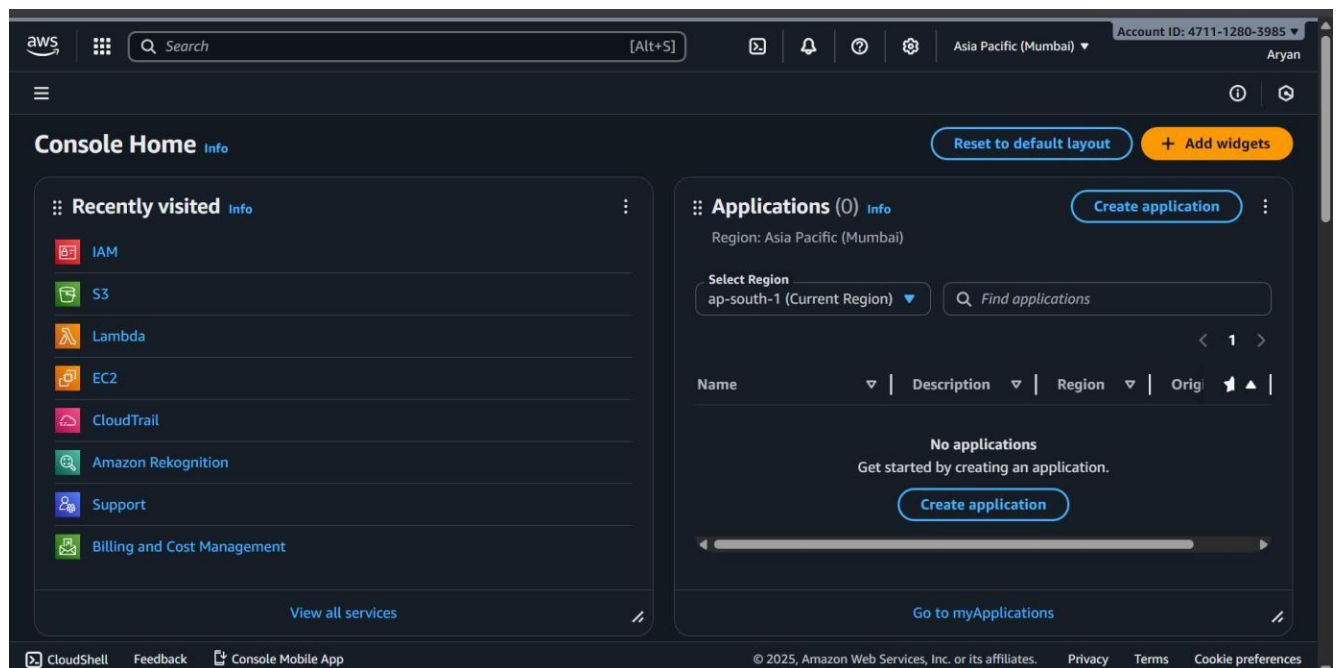


Practical :-6

Objective:- To secure an AWS IAM user account by enabling Multi-Factor Authentication (MFA), adding an extra verification step to prevent unauthorized access.

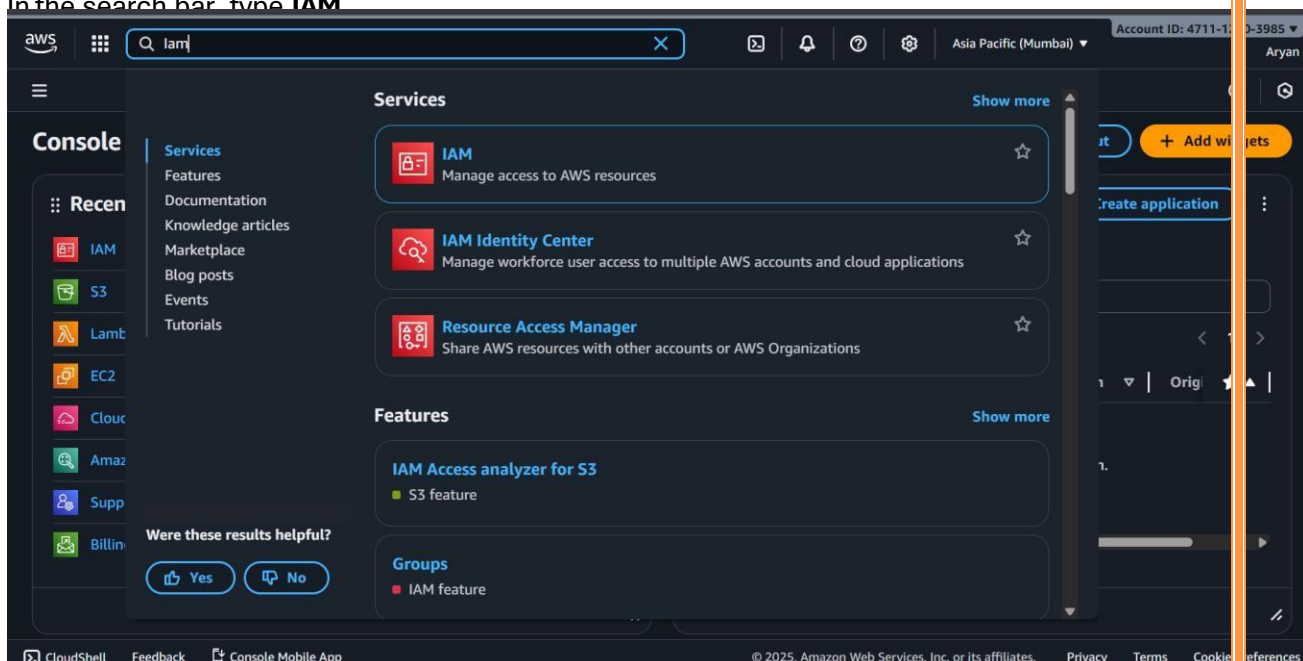
Step 1:- Sign in to AWS Management Console

- Log in using your **root account** or **IAM admin user**.
- Open the console.



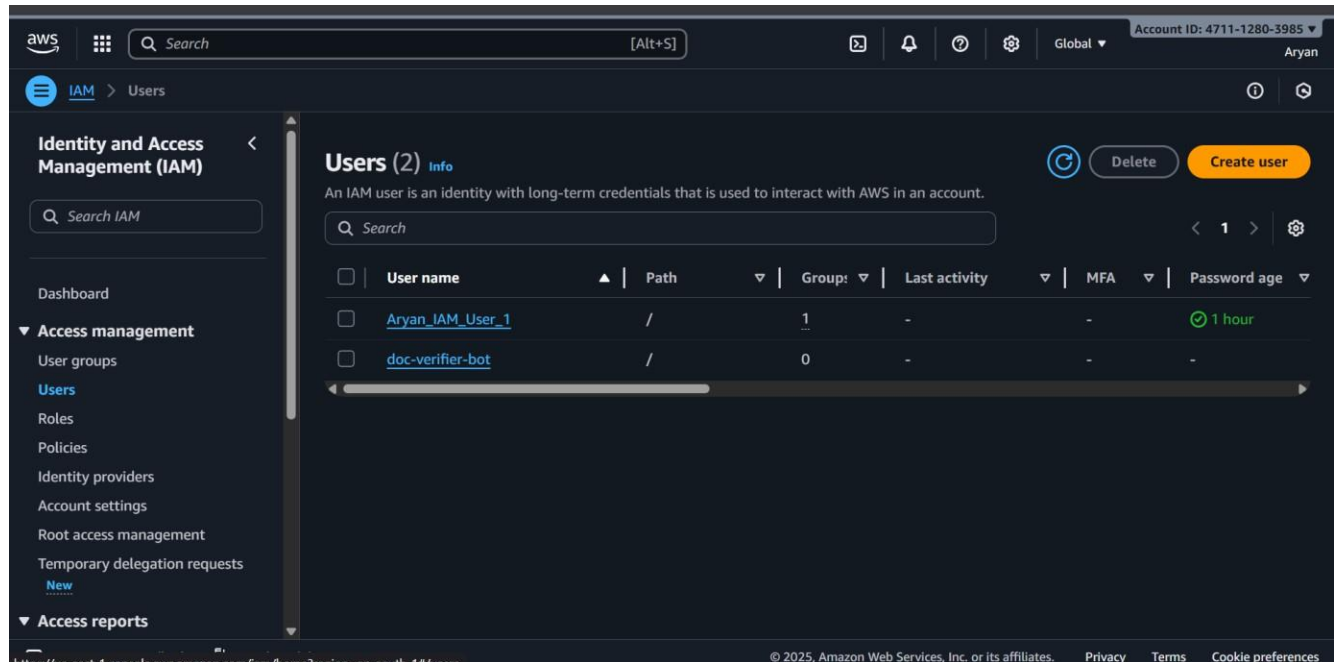
Step 2:- Go to IAM Service

- In the search bar, type **IAM**



Click IAM (Identity and Access Management).

Step 3:-Open “Users”

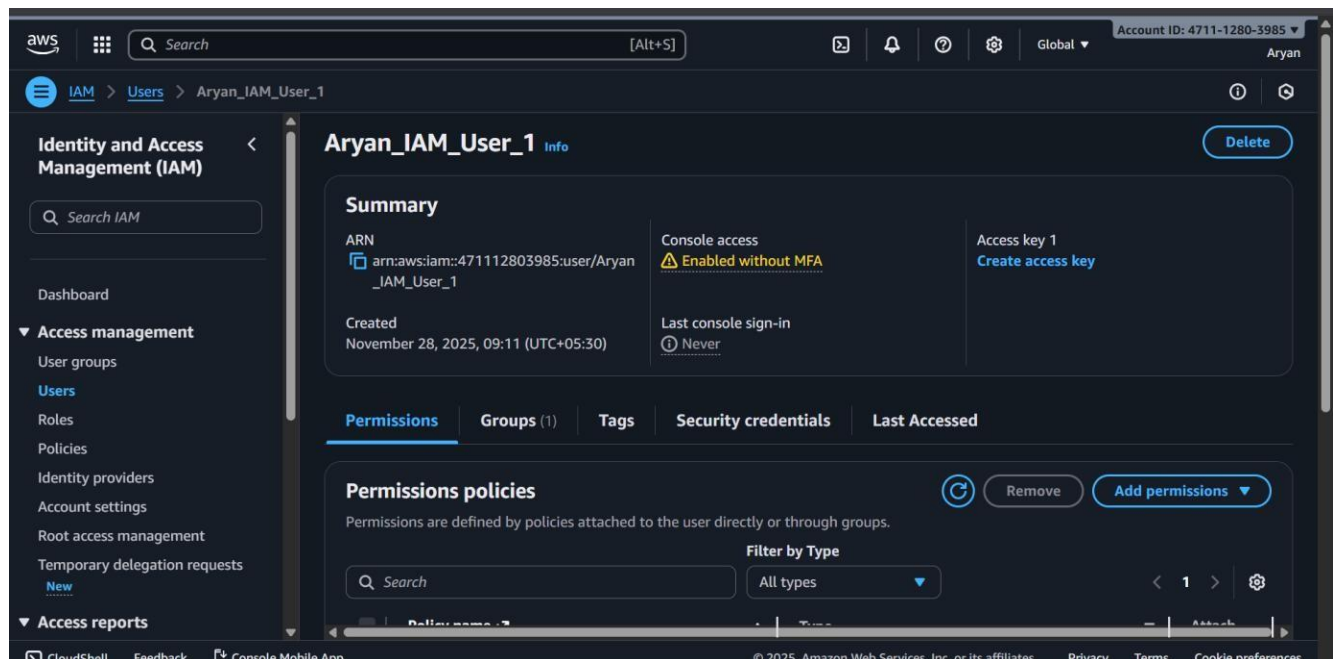


- In the left navigation panel, select **Users**.

Click the **username** for which you want to enable MFA.

Step 4:-Go to the “Security Credentials” Tab

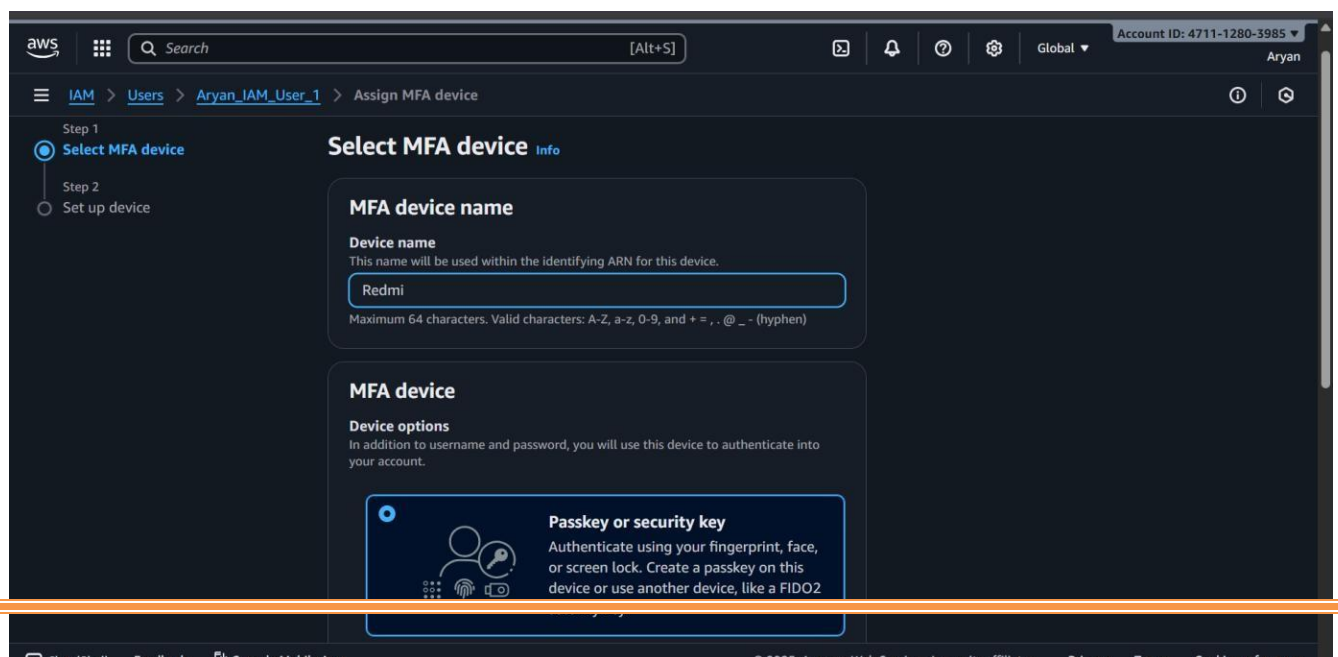
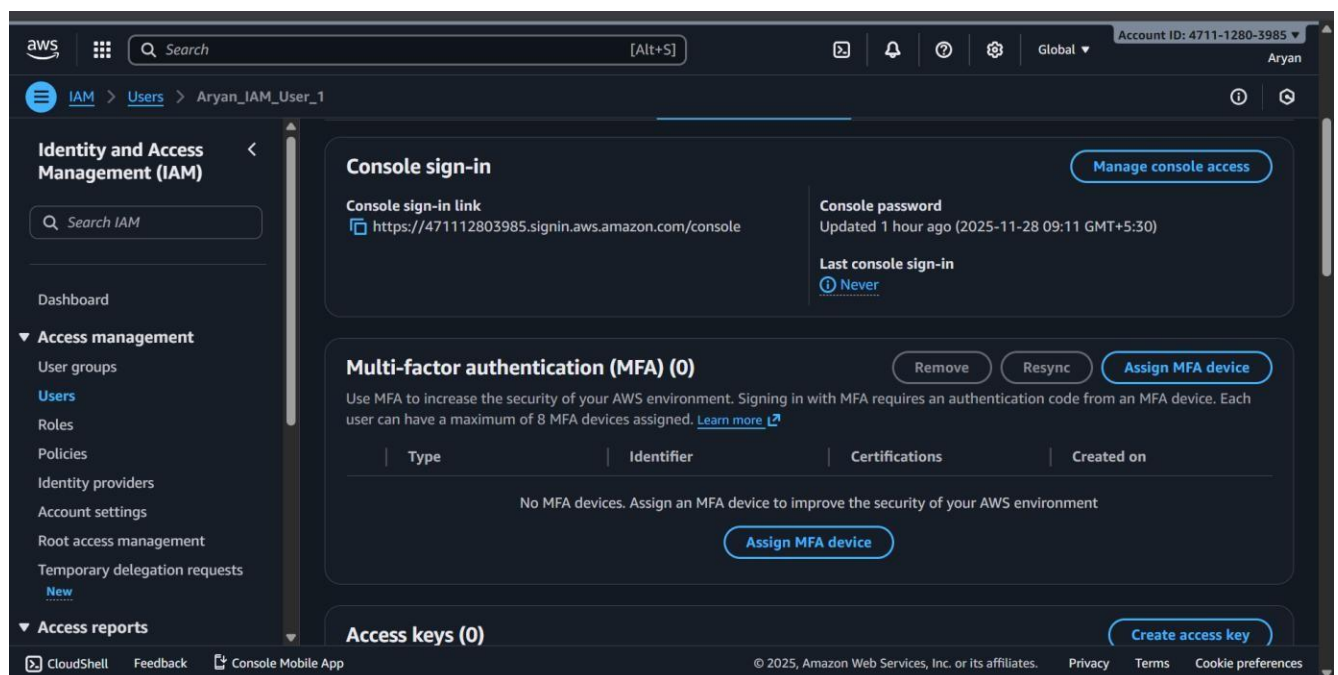
- After opening the user’s profile, click on **Security credentials**.
- Scroll down to the section **Multi-Factor Authentication (MFA)**.



Step 5:-Click “Assign MFA Device”

- A popup will open with 3 options:
 1. **FIDO2 Security Key**
 2. **Authenticator App** (Google Authenticator / Authy / Microsoft Authenticator)
 3. **Hardware TOTP Device**

For most labs, choose **Authenticator App**.



Step 6:-Select “Authenticator App” and Continue

- Click **Continue**.
- AWS will show a **QR code** for MFA enrollment.

Step 7:-Open your Authenticator App

You can use any app:

- Google Authenticator
- Authy
- Microsoft Authenticator
- LastPass Authenticator

Click **Add Account** → **Scan QR Code**.

The app will generate a **6-digit one-time password (OTP)** that refreshes every 30 seconds.

2

Open your authenticator app, choose **Show QR code** on this page, then use the app to scan the code. Alternatively, you can type a secret key. [Show secret key](#)

3

Type two consecutive MFA codes below

Enter a code from your virtual app below

MFA Code 1

Wait 30 seconds, and enter a second code entry.

MFA Code 2

Cancel Previous Add MFA

Activate Windows
Go to Settings to activate Windows

Step 8:-MFA Successfully Enabled

“MFA device assigned successfully”

The screenshot displays the AWS IAM console interface. At the top, a green notification banner states "MFA device assigned" with a checkmark icon, explaining that up to 8 MFA devices can be registered and that only one is needed for sign-in. The breadcrumb navigation shows the path: IAM > Users > Aryan_IAM_User_1. The left sidebar contains the "Identity and Access Management (IAM)" menu with options like Dashboard, Access management (Users, Roles, Policies, etc.), and Access reports. The main content area is titled "Aryan_IAM_User_1" and includes a "Delete" button. Below the title is a "Summary" section with a table of user details:

Summary		
ARN	Console access	Access key 1
arn:aws:iam::471112803985:user/Aryan_IAM_User_1	Enabled with MFA	Create access key
Created	Last console sign-in	
November 28, 2025, 09:11 (UTC+05:30)	Never	

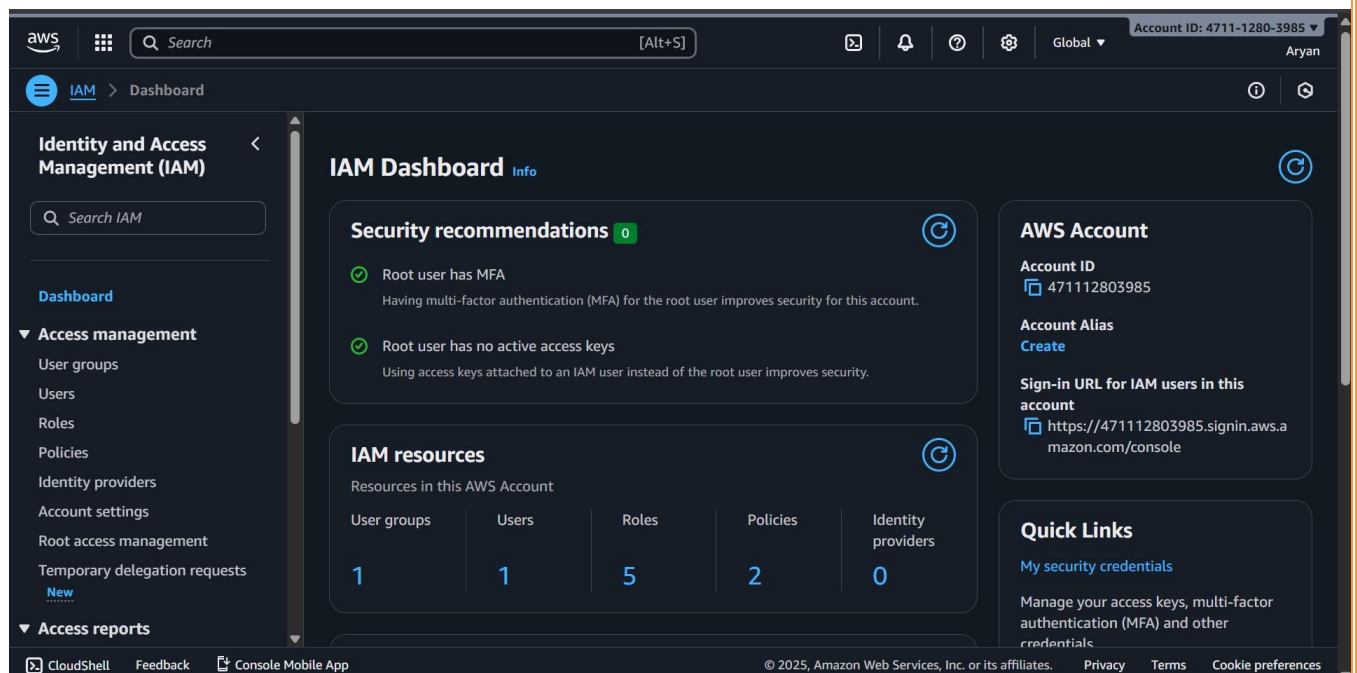
Below the summary is a tabbed interface with "Security credentials" selected. It shows a "Console sign-in" section with a "Manage console access" button. The footer includes links for CloudShell, Feedback, Console Mobile App, and copyright information for Amazon Web Services, Inc. or its affiliates.

Practical:-7

Objective: To create a user group in AWS IAM in order to manage permissions collectively for multiple users having similar roles or responsibilities.

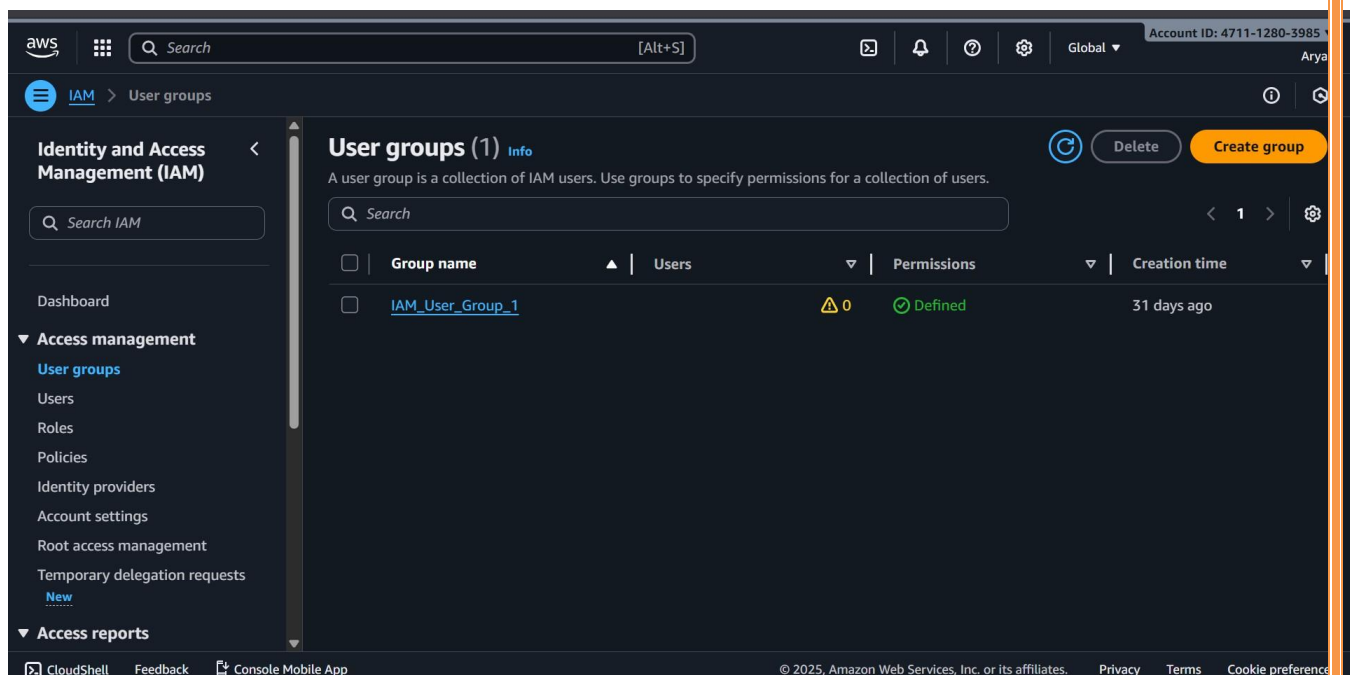
Step1:- Open the IAM Service:

In the search bar at the top of the console, type **IAM**, then select **Identity and Access Management** from the results.



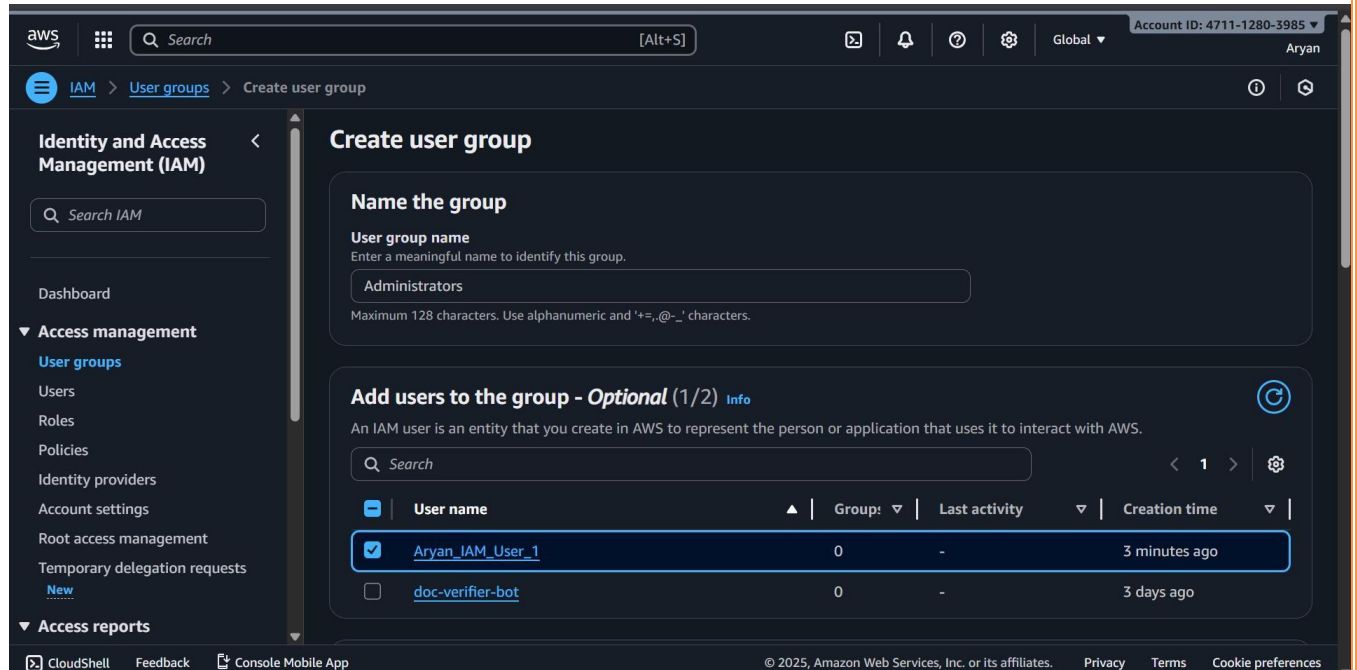
Step 2:- Go to User Groups Section:

In the left-hand sidebar, click on **User groups**.



Step 3:- Click on “Create group”:

On the User Groups page, click the “**Create group**” button to start creating a new group



Step 4:- Enter Group Name:

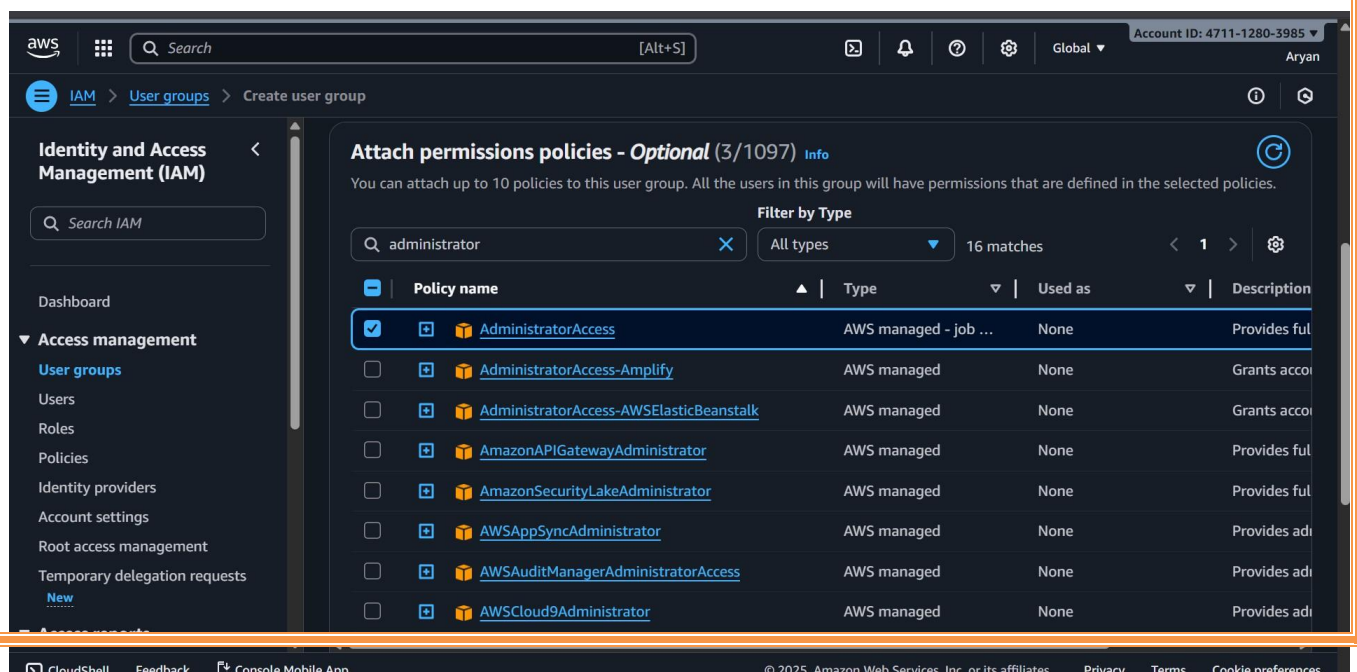
Type a **unique name** for your group (for example, *Developers*, *Admins*, or *ReadOnlyUsers*).
For Example, here Administrators.

Step 5:- Attach Permissions Policies (Optional):

You can choose policies to attach to this group, such as:

- AmazonS3FullAccess
- AmazonEC2ReadOnlyAccess
- AdministratorAccess

If you want to add permissions later, you can **skip this step** and click **Next**.



Step 6:- Add Users to the Group (Optional):

You can select existing IAM users to include in this group now, or you can add users later after creating the group.

Step 7:- Review and Create Group:

Review the group details and attached policies, then click **Create group**.

aws [Search] [Alt+S] Global Account ID: 4711-1280-3985 Aryan

Identity and Access Management (IAM) Search IAM

Dashboard

Access management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests
- New

Access reports

Administrators user group created. View group

User groups (2) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

☐	Group name	Users	Permissions	Creation time
☐	Administrators	1	Defined	Now
☐	IAM_User_Group_1	0	Defined	31 days ago

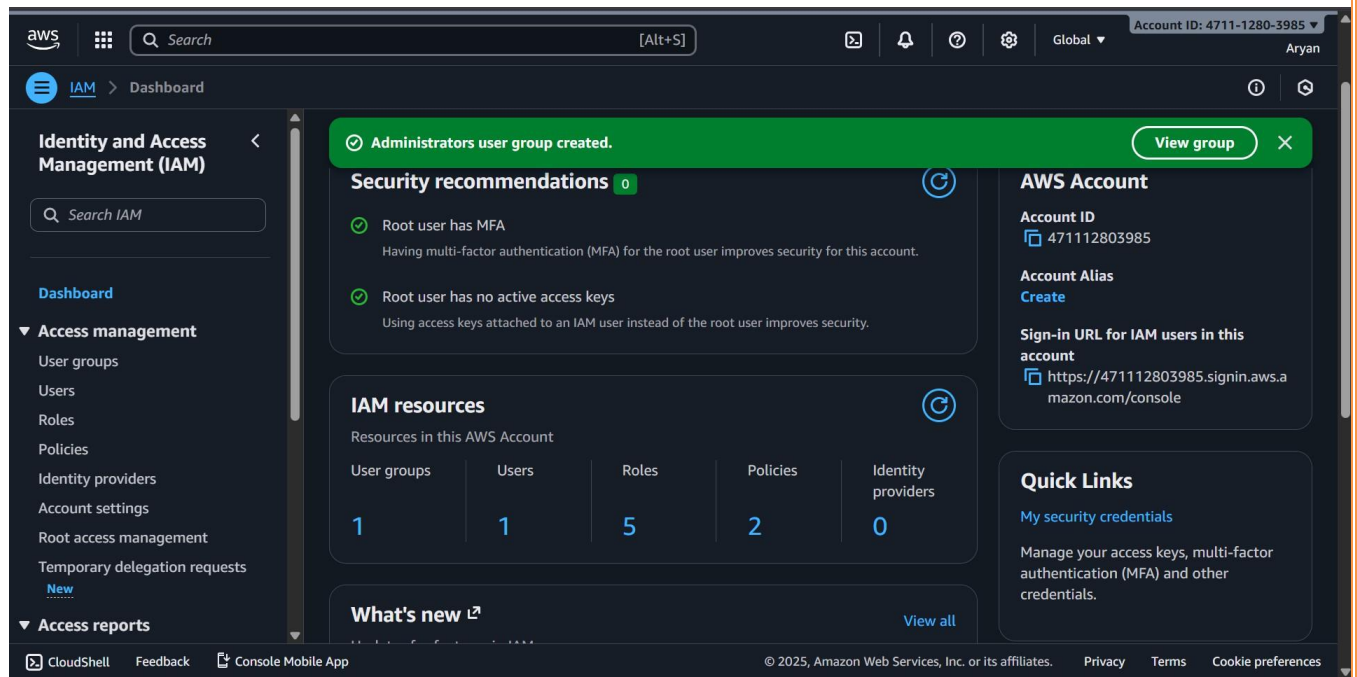
CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Practical:-8

Objective:- To create a security role in AWS IAM that allows AWS services or users to securely access specific AWS resources with defined permissions, ensuring controlled and temporary access without sharing long-term credentials.

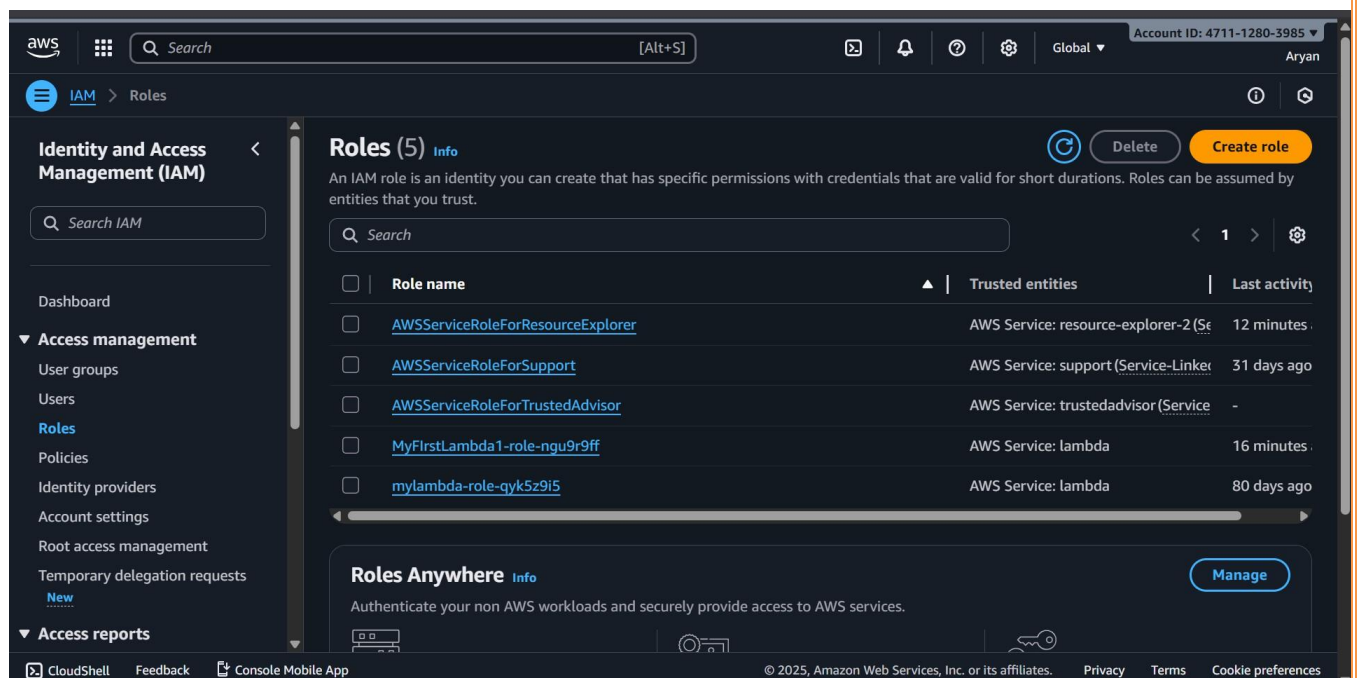
Step 1:- Open the IAM Service:

In the search bar at the top of the console, type **IAM**, then select **Identity and Access Management** from the results.



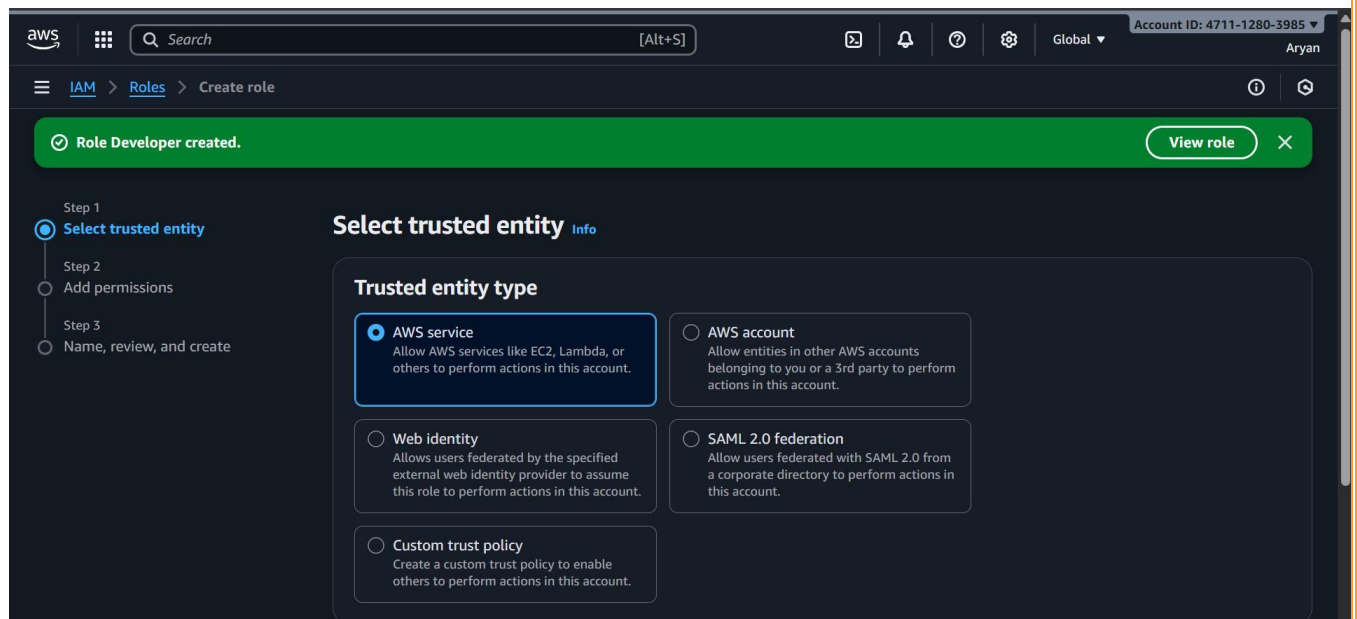
Step 2:- Go to Roles Section:

In the left-hand navigation pane, click on **Roles**.



Step 3:- Click on “Create role”:

On the Roles page, click the “Create role” button to start the process.

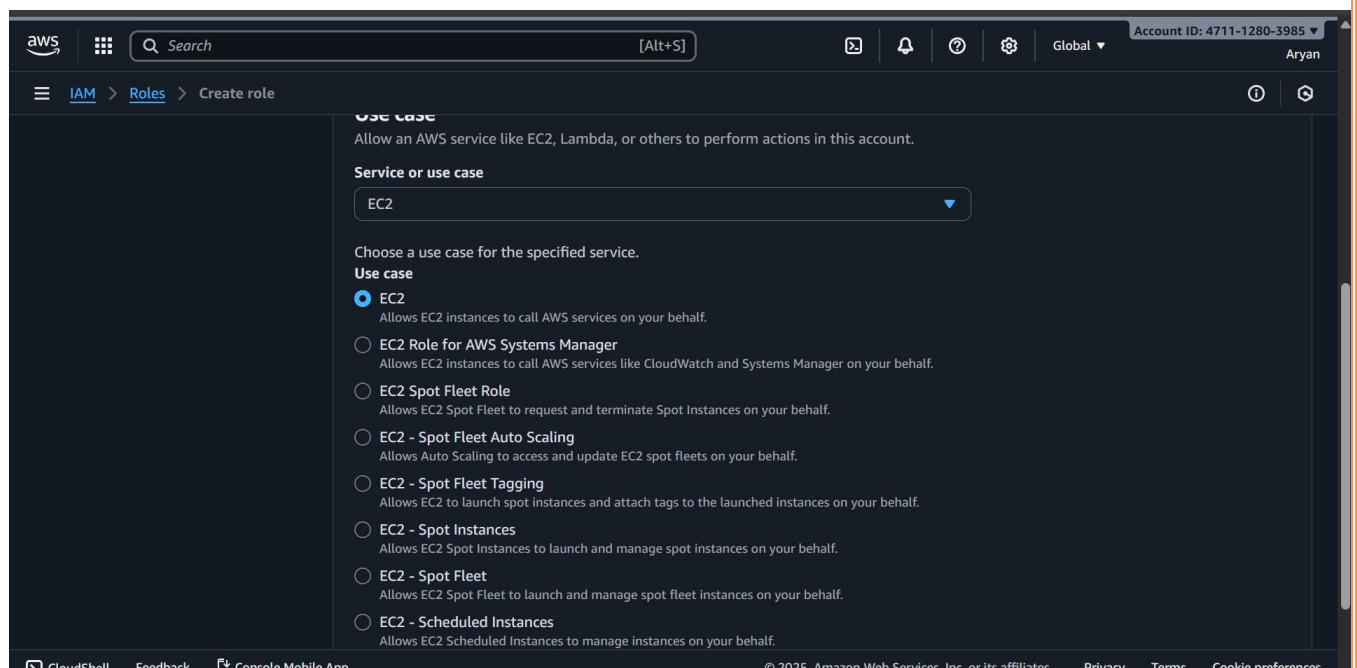


Step 4:- Select Trusted Entity Type:

Choose who will use the role, such as:

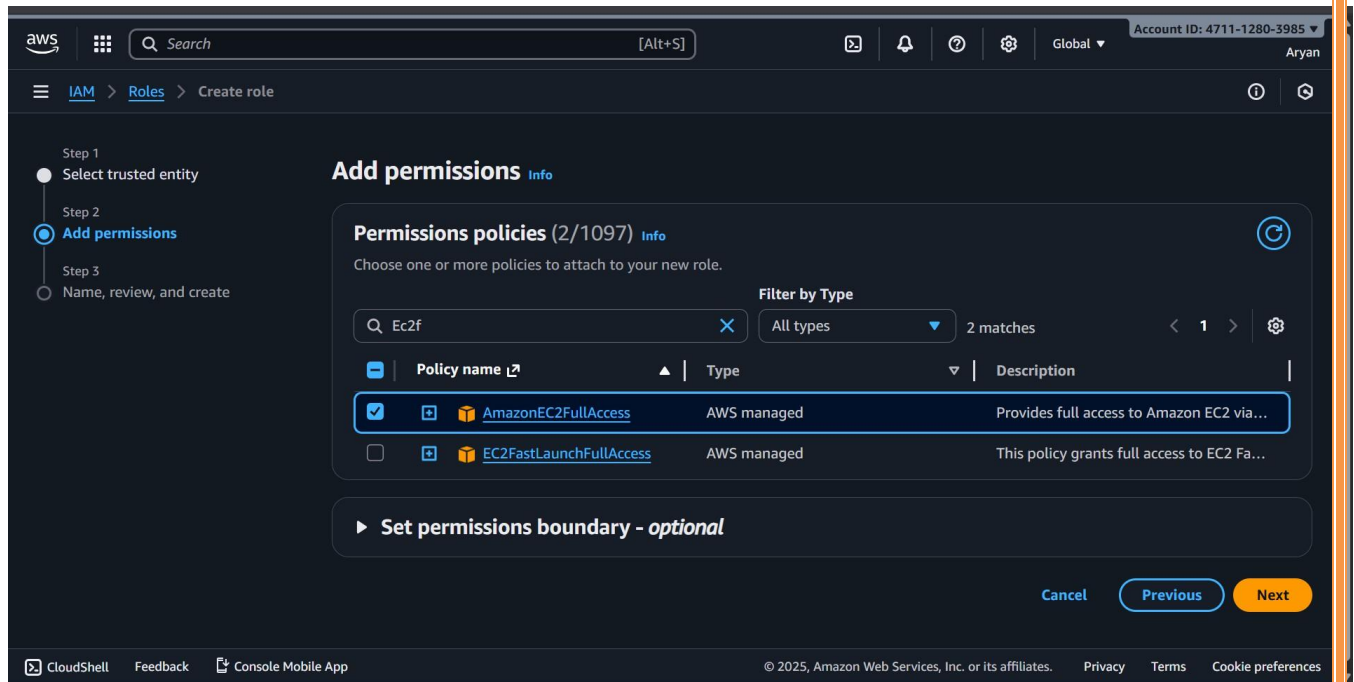
- **AWS Service** (e.g., EC2, Lambda)
- **Another AWS Account**
- **Web Identity** or **SAML 2.0 Federation**

Click **Next** after selecting the appropriate option.



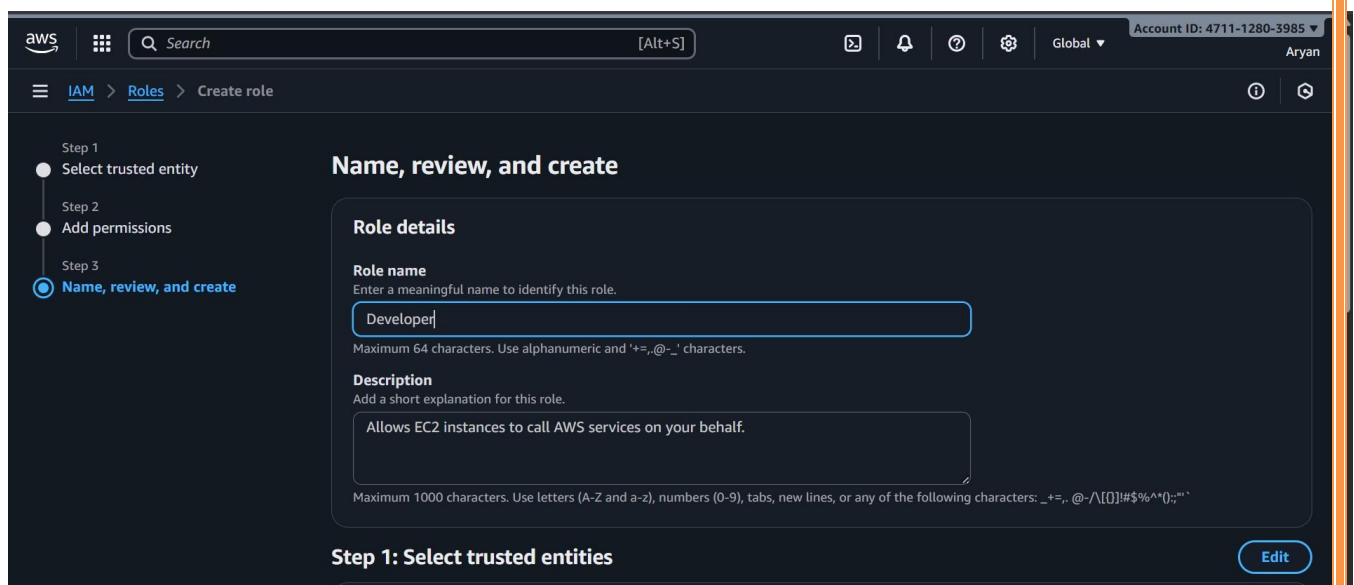
Step 5:-Attach Permissions Policies:

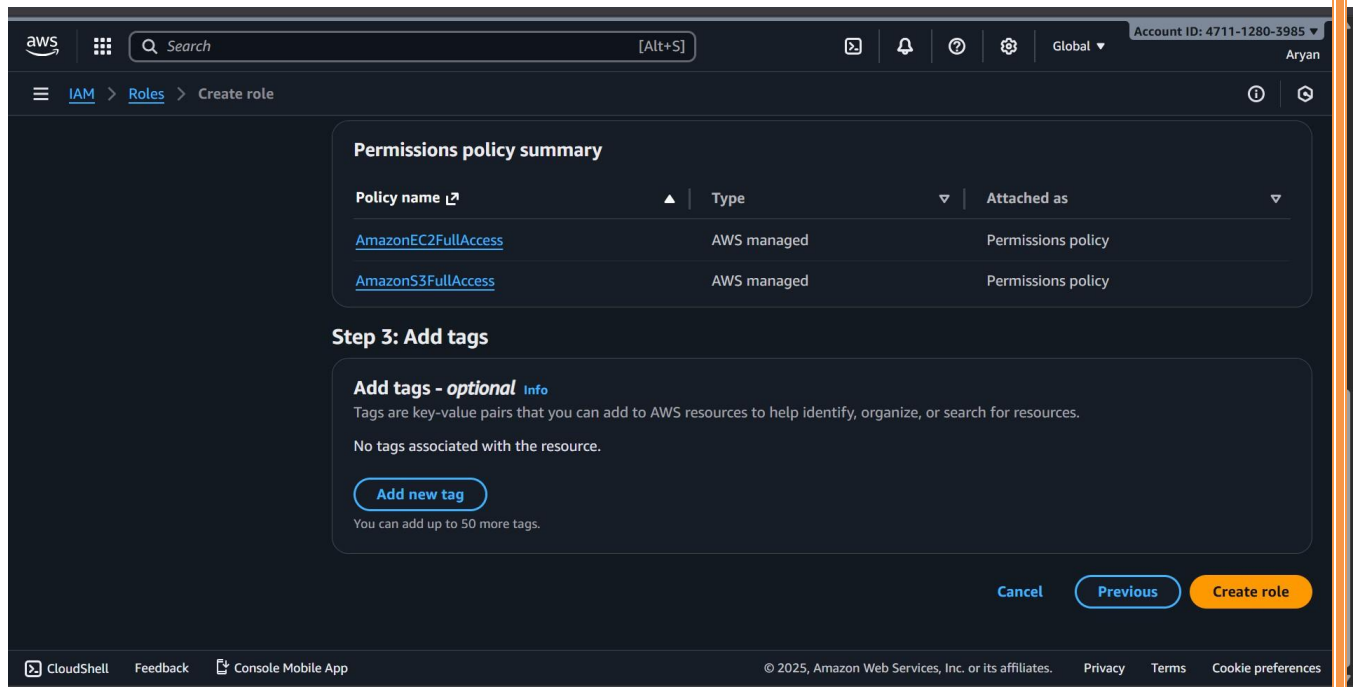
Select the **permissions policies** that define what actions the role can perform (for example, AmazonS3FullAccess or AmazonEC2FullAccess).



Step 6:- Name and Review the Role:

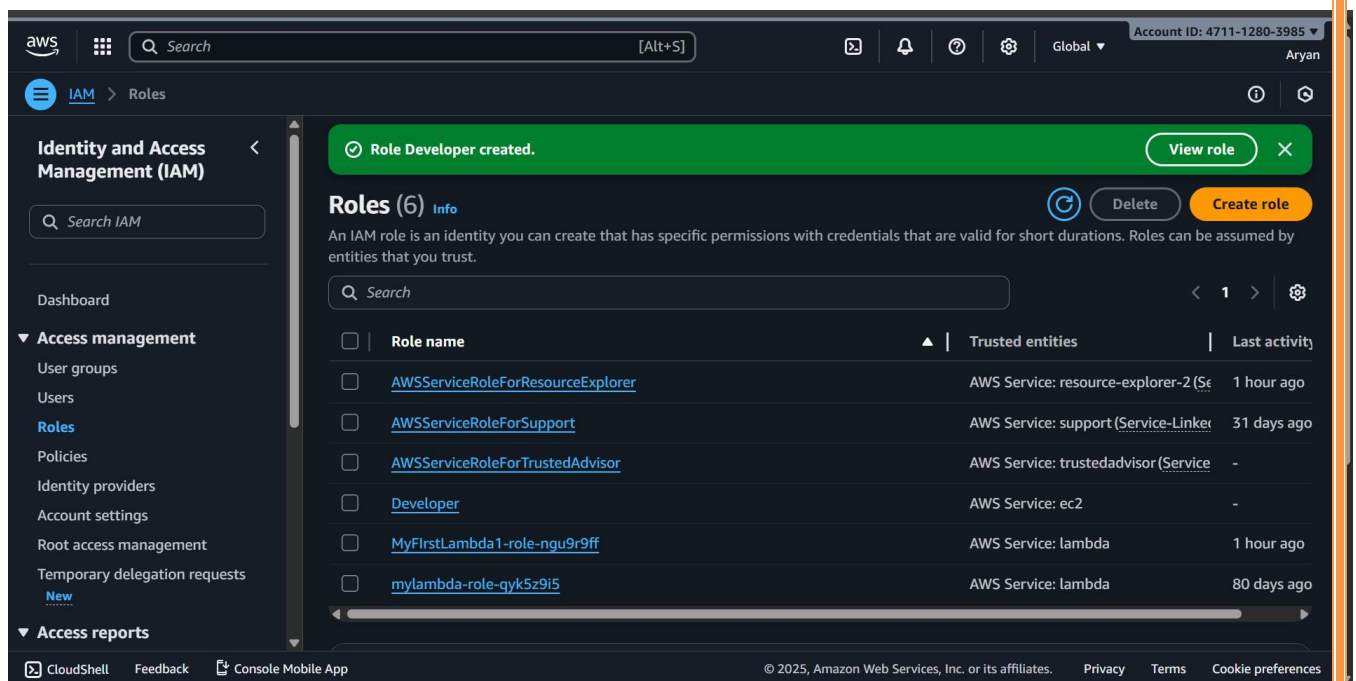
Enter a **role name** (for example, *EC2SecurityRole* or *LambdaAccessRole*) and review all selected settings.





Step7:- Create the Role:

Click **Create role** to finish.

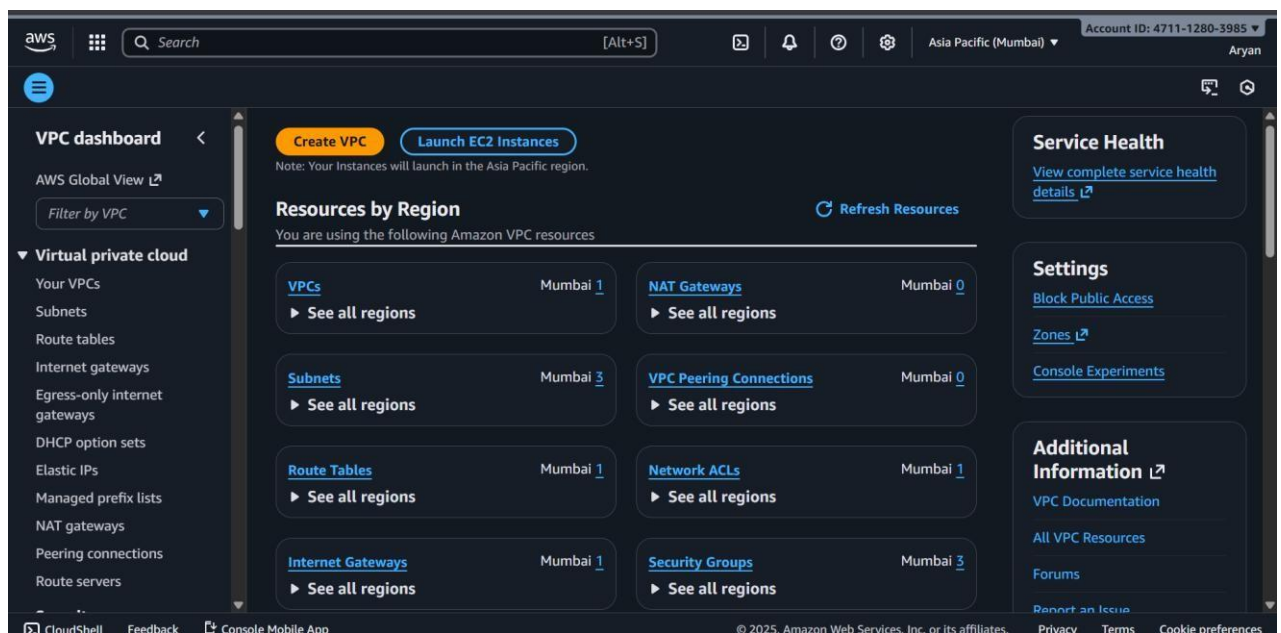


Practical :- 9

Objective:- To understand how to design and create a Virtual Private Cloud (VPC) in AWS.

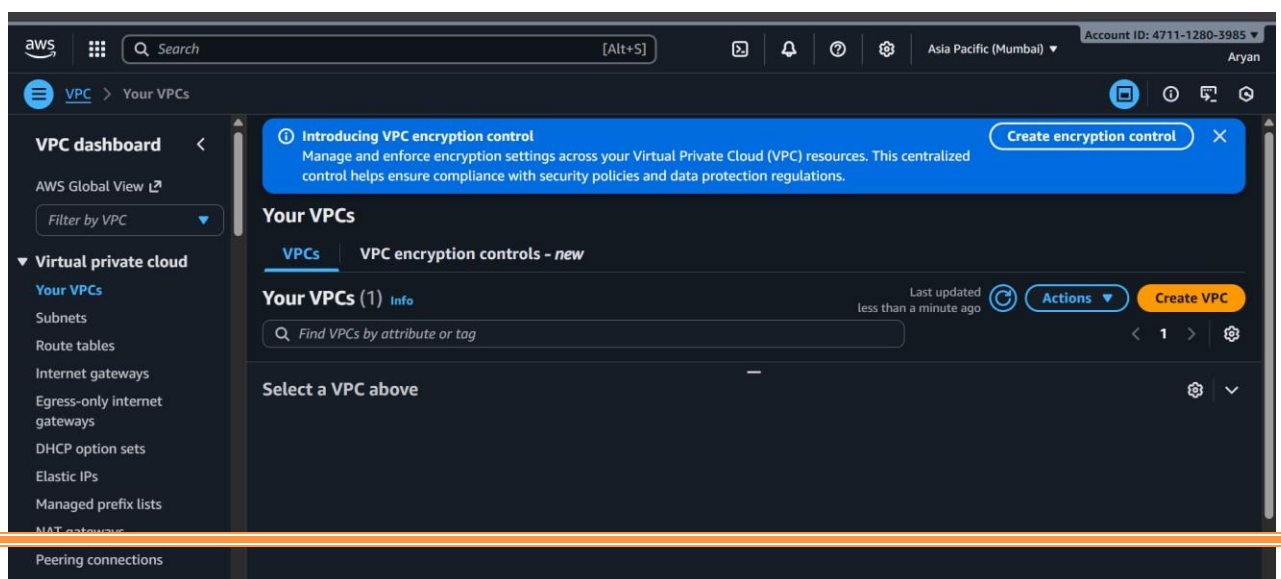
Step1. Open VPC Dashboard

- Sign in to AWS Console.
- Search **VPC** in the services search bar.
- Click **VPC Dashboard**.



Step2. Start Creating a VPC

- On the left menu, select **Your VPCs**.
- Click **Create VPC**.

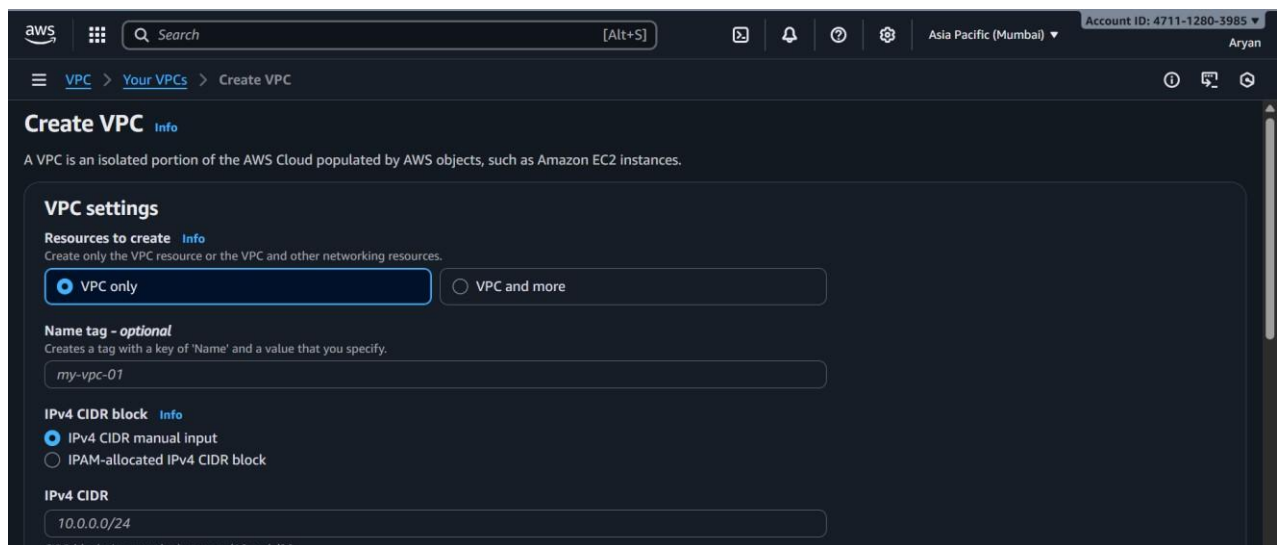


Step3. Select VPC Creation Method

You will get two options:

1. **VPC Only** → Create VPC manually
2. **VPC and more** → Automatically create VPC with subnets, IGW, route tables, etc.

Choose **VPC Only** for full control.



Step 4. Configure VPC Settings

Fill the form:

(a) Name tag

- Example: MyVPC

(b) IPv4 CIDR Block

- Example: 10.0.0.0/16

(c) IPv6 CIDR Block

- Choose **No IPv6 CIDR block** (optional)

(d) Tenancy

- Default (recommended)

Click **Create VPC**.

The screenshot shows the 'Create VPC' page in the AWS Management Console. The page is titled 'VPC settings' and includes a 'Resources to create' section with two radio buttons: 'VPC only' (selected) and 'VPC and more'. Below this is a 'Name tag - optional' section with a text input field containing 'my-VPC-1'. The 'IPv4 CIDR block' section has two radio buttons: 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block'. The 'IPv4 CIDR' text input field contains '10.0.0.0/16'. The 'IPv6 CIDR block' section has four radio buttons: 'No IPv6 CIDR block' (selected), 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me'.

Step 5. Create Subnets

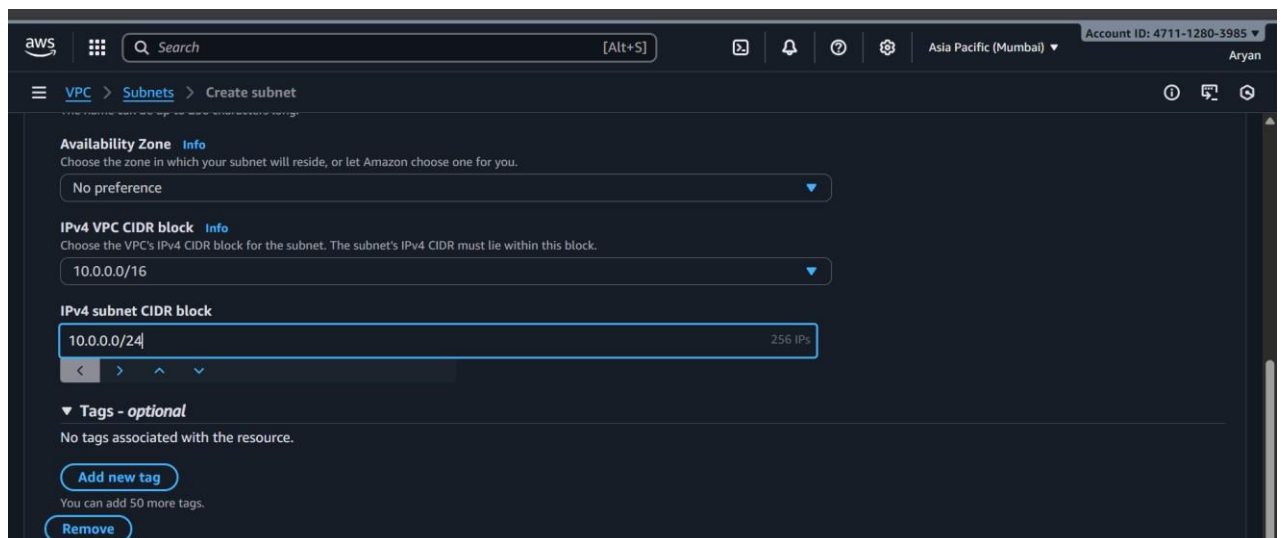
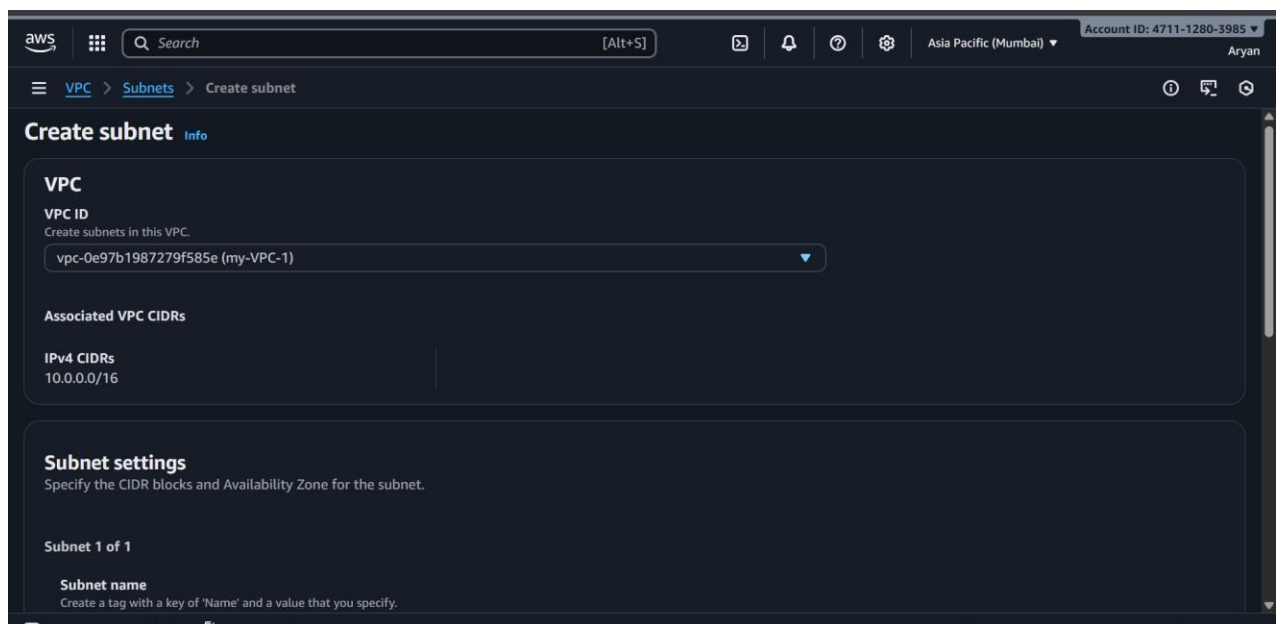
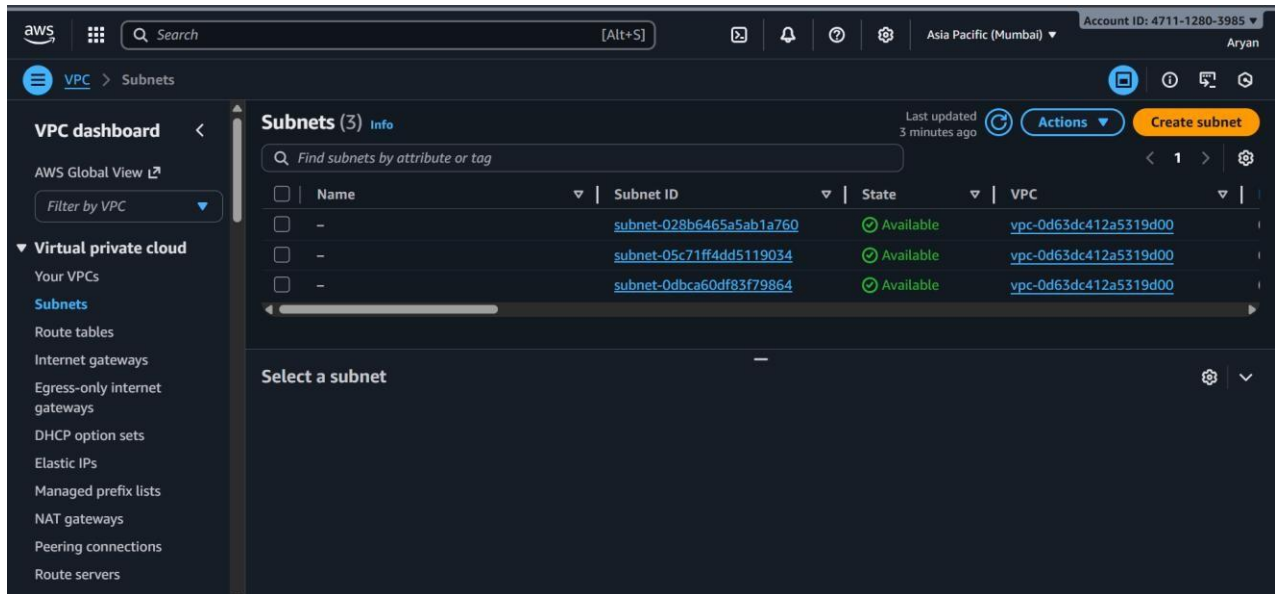
- On the left side, click **Subnets** → **Create subnet**.
- Select your **VPC**.
- Add at least **two subnets**:
 - Public → 10.0.1.0/24
 - Private → 10.0.2.0/24
- Choose different **Availability Zones**.

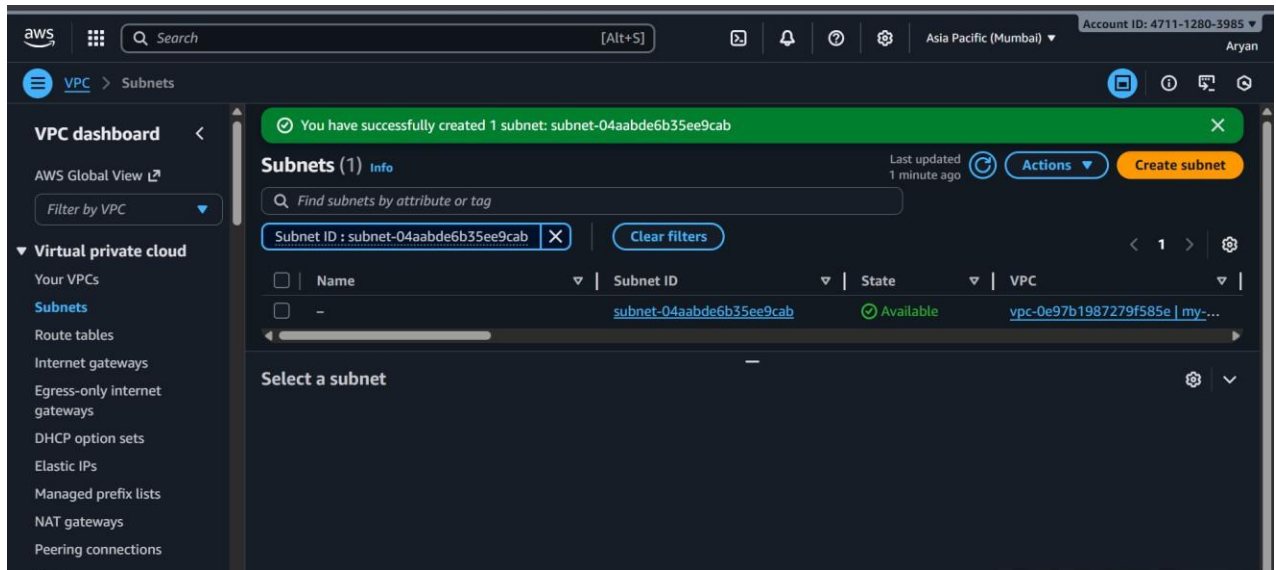
Click **Create Subnet**.

The screenshot shows the 'VPC dashboard' in the AWS Management Console. A green notification bar at the top states 'You successfully created vpc-0e97b1987279f585e / my-VPC-1'. The main content area displays the details for 'vpc-0e97b1987279f585e / my-VPC-1'. The details are organized into a grid with the following sections:

- Details**
 - VPC ID: vpc-0e97b1987279f585e
 - State: Available
 - DNS resolution: Enabled
 - Main network ACL: acl-0662499a3c9ed0ed0
 - IPv6 CIDR (Network border group): -
 - Encryption control ID: -
- Tenancy**
 - Tenancy: default
 - Default VPC: No
 - Network Address Usage metrics: Disabled
 - Encryption control mode: -
- Block Public Access**
 - Block Public Access: Off
- DHCP option set**
 - DHCP option set: dopt-0646f9c3427a835b8
- Route 53 Resolver DNS Firewall rule groups**
 - Route 53 Resolver DNS Firewall rule groups: -
- DNS hostnames**
 - DNS hostnames: Disabled
- Main route table**
 - Main route table: rtb-07fa7122b397cf010
- IPv6 pool**
 - IPv6 pool: -
- Owner ID**
 - Owner ID: 471112803985

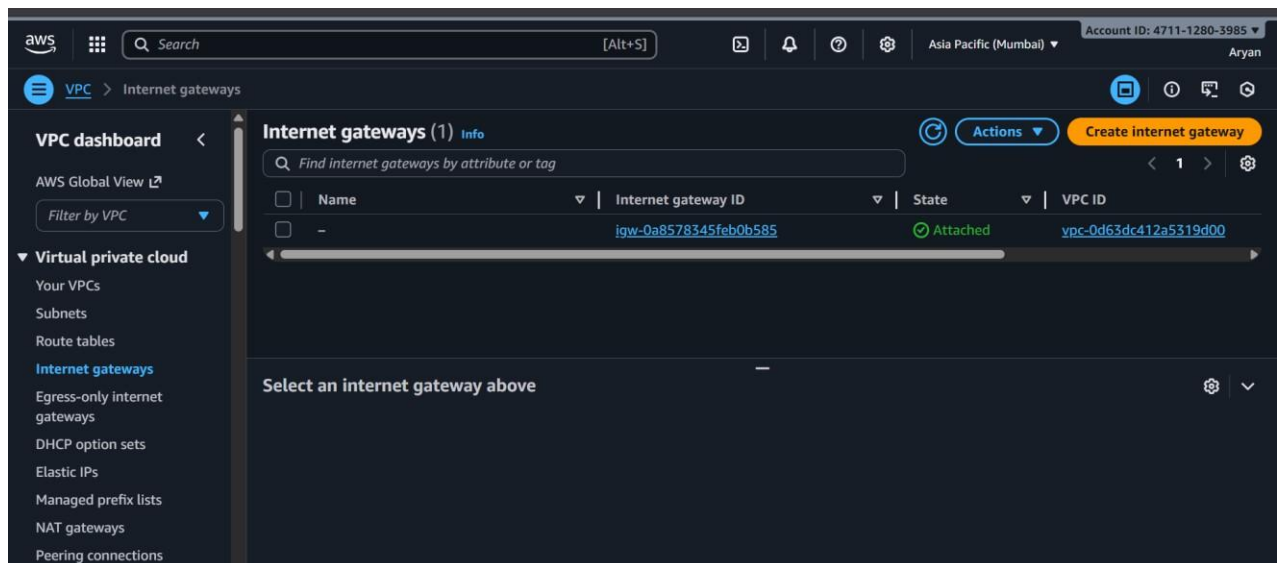
At the bottom of the details section, there are tabs for 'Resource map', 'CIDRs', 'Flow logs', 'Tags', and 'Integrations'. The 'Resource map' tab is currently selected.





Step 6. Create an Internet Gateway (IGW)

- Click **Internet Gateways** → **Create IGW**.
- Name it e.g., MyIGW.
- After creation, select it → click **Attach to VPC** → choose your **VPC**.



aws

Search

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Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Internet gateways

Create internet gateway

Create internet gateway

Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Creates a tag with a key of 'Name' and a value that you specify.

myIGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

X

Value - optional

Q myIGW

X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Internet gateways

igw-0079fabea618e738a

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

The following internet gateway was created: igw-0079fabea618e738a - myIGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

igw-0079fabea618e738a / myIGW

Actions

Details

Info

Internet gateway ID

igw-0079fabea618e738a

State

Detached

VPC ID

-

Owner

471112803985

Tags (1)

Search tags

Key

Value

Name

myIGW

Manage tags

1

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Internet gateways

Attach to VPC (igw-0079fabea618e738a)

Attach to VPC (igw-0079fabea618e738a)

Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

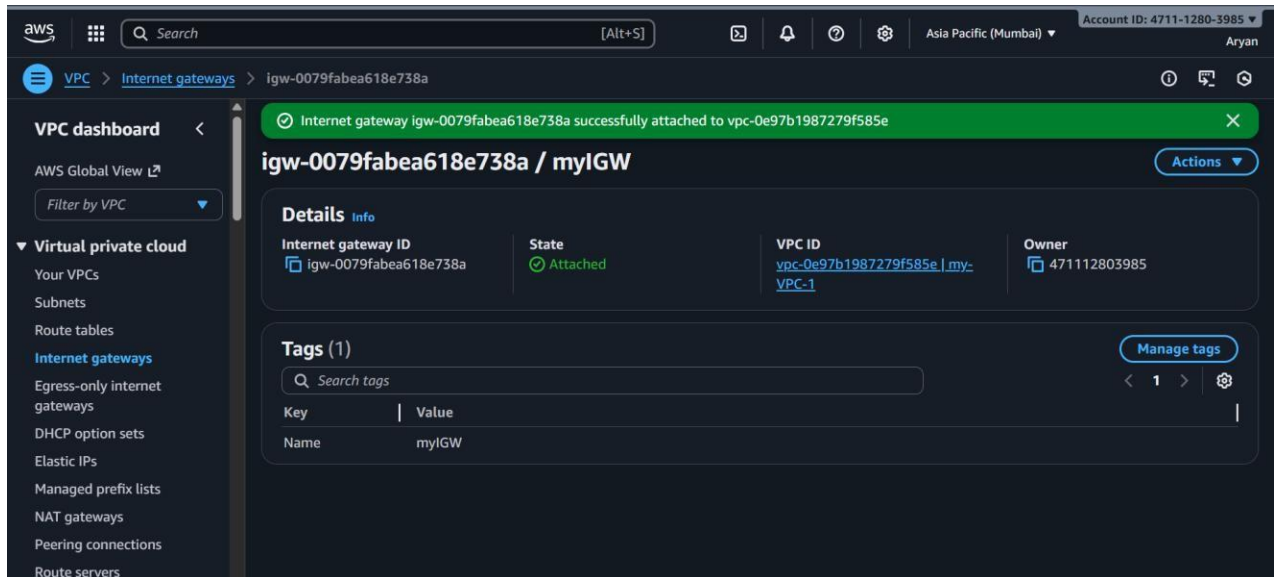
Q vpc-0e97b1987279f585d

X

AWS Command Line Interface command

Cancel

Attach internet gateway



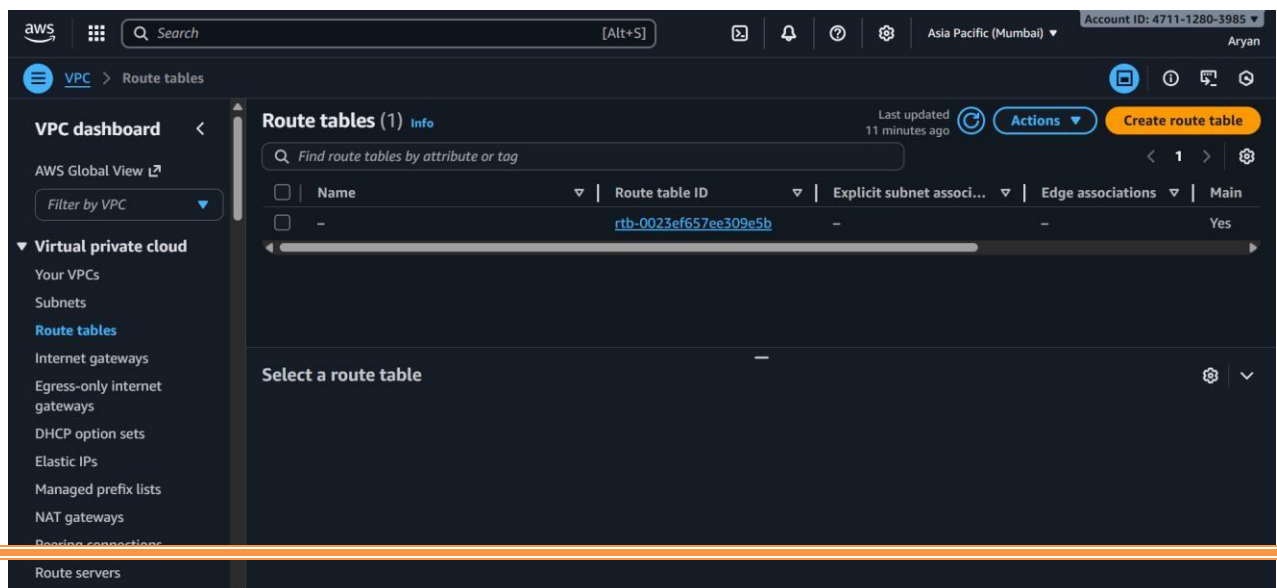
Step 7. Configure Route Tables

Public Route Table

- Go to **Route Tables** → Create route table.
- Name: PublicRT.
- Select your VPC and create.
- Go to **Routes** → Edit → Add route:
 - Destination: 0.0.0.0/0
 - Target: **Internet Gateway**
- Go to **Subnet Associations** → Associate with **public subnet**.

Private Route Table

- Create another route table named PrivateRT.
- Associate it with **private subnet** (no internet route).



aws

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Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

Create route table

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Public-RT

VPC
The VPC to use for this route table.
vpc-0e97b1987279f585e (my-VPC-1)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
Public-RT

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

rtb-0b1c88c61c45abea4

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
	local			

Add route

Cancel

Preview

Save changes

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

rtb-0b1c88c61c45abea4

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
	local			
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	igw-			

Add route

Cancel

Preview

Save changes

Remove

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

rtb-0b1c88c61c45abea4

VPC dashboard

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Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Updated routes for rtb-0b1c88c61c45abea4 / Public-RT successfully

Details

rtb-0b1c88c61c45abea4 / Public-RT

Actions

Details

Info

Route table ID

rtb-0b1c88c61c45abea4

Main

No

Explicit subnet associations

-

Edge associations

-

VPC

vpc-0e97b1987279f585e | my-VPC-1

Owner ID

471112803985

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

Destination

Target

Status

Propagated

Route Origin

0.0.0.0/0

lgw-0079fabea618e7...

Active

No

Create Route

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

Create route table

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Private-RT

VPC

The VPC to use for this route table.

vpc-0e97b1987279f585e (my-VPC-1)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

Private-RT

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Account ID: 4711-1280-3985

Aryan

VPC

Route tables

rtb-0e296cd139a06a61b

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Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Route table rtb-0e296cd139a06a61b | Private-RT was created successfully.

rtb-0e296cd139a06a61b / Private-RT

Actions

Details

Info

Route table ID

rtb-0e296cd139a06a61b

Main

No

Explicit subnet associations

-

Edge associations

-

VPC

vpc-0e97b1987279f585e | my-VPC-1

Owner ID

471112803985

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

Destination

Target

Status

Propagated

Route Origin

10.0.0.0/16

local

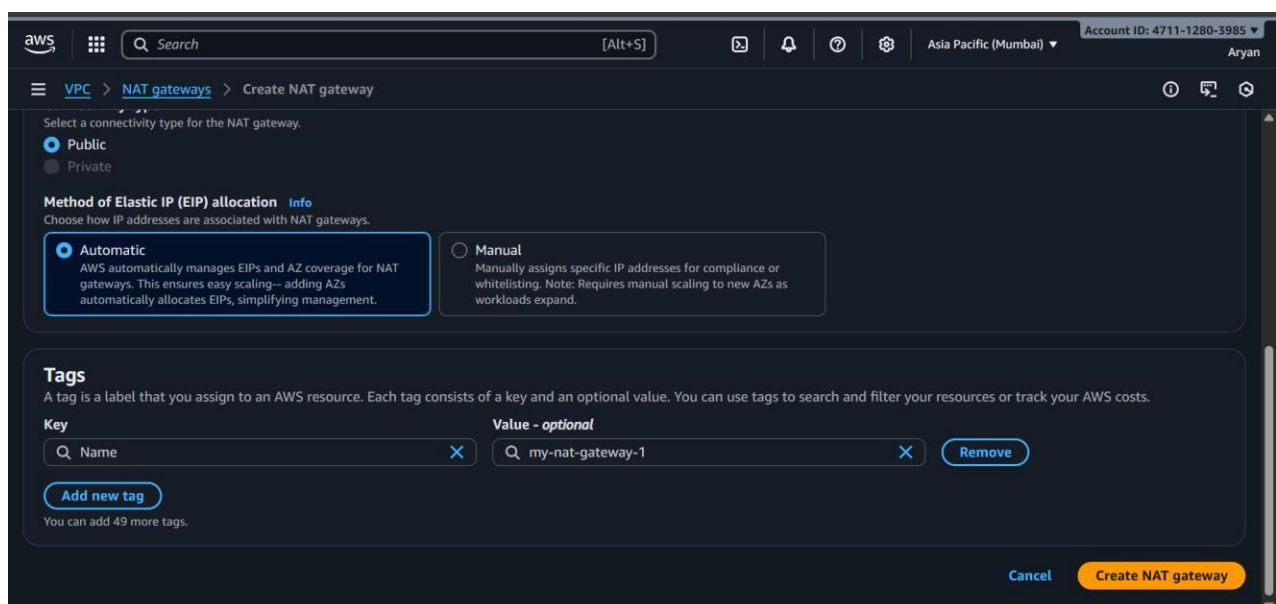
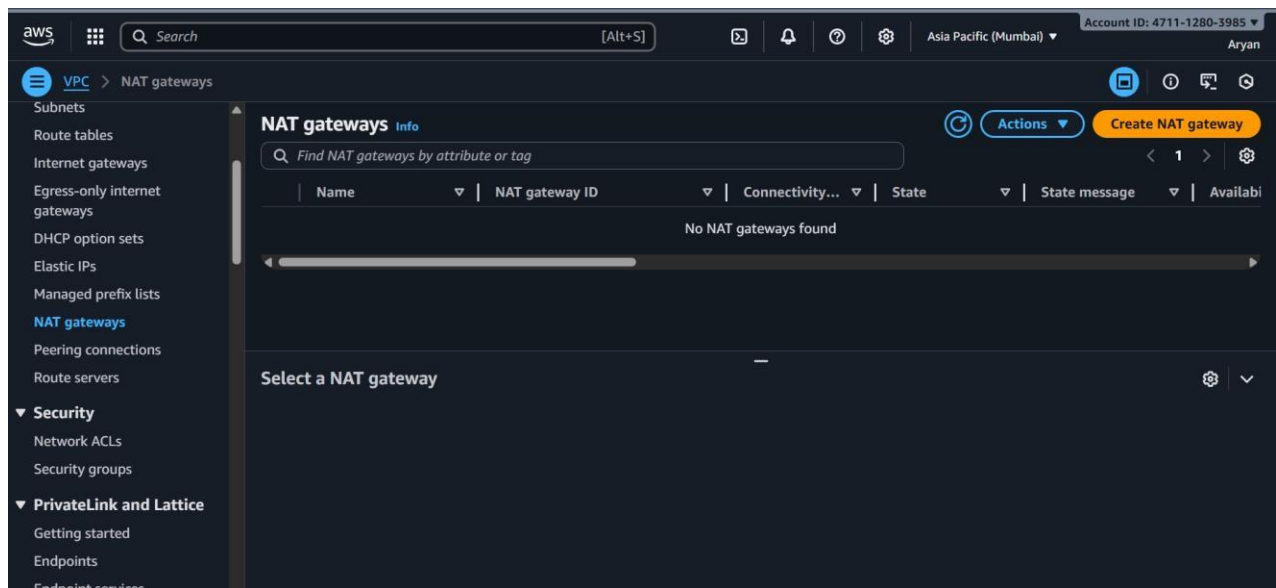
Active

No

Create Route Table

Step 8. Create a NAT Gateway (Optional for private subnet internet)

- Go to **NAT Gateways** → Create NAT Gateway.
- Select:
 - Subnet: **Public subnet**
 - Allocate Elastic IP
- Update **PrivateRT** → **Routes** to:
 - Destination: 0.0.0.0/0
 - Target: **NAT Gateway**



aws

Search

[Alt+S]

Account ID: 4711-1280-3985

Asia Pacific (Mumbai)

Aryan

VPC > NAT gateways > nat-1017dbc3d228e408f

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

NAT gateway nat-1017dbc3d228e408f | my-nat-gateway-1 was created successfully.

nat-1017dbc3d228e408f / my-nat-gateway-1

Actions

Details

NAT gateway ID

nat-1017dbc3d228e408f

NAT gateway ARN

arn:aws:ec2:ap-south-1:471112803985:natgateway/nat-1017dbc3d228e408f

VPC

vpc-0e97b1987279f585e / my-VPC-1

Availability mode

Regional

Connectivity type

Public

Method of EIP allocation

Automatic

State

Pending

Created

Friday, November 28, 2025 at 11:02:29 GMT+5:30

State message

Info

Deleted

IP addresses

Monitoring

Flow logs

Tags

Associated IP addresses

Search

Edit IP address associations

< 1 >